

CLD200

Building Side-by-Side Extensions on SAP BTP

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

Course Version: 23
Course Duration: 2 Days
Material Number: 50167129

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation



Demonstration



Procedure



Warning or Caution



Hint



Related or Additional Information



Facilitated Discussion



User interface control

Example text

Window title

Example text

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Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

- Application Consultant
- Development Consultant
- Super / Key / Power User
- Business Process Architect
- Business Process Owner/Team Lead/Power User
- Developer
- Enterprise Architect
- Solution Architect

UNIT 1

Discovering the SAP Cloud Application Programming Model (CAP)

Lesson 1

Identifying the Need for Side-By-Side Extensibility

3

Lesson 2

Exploring the SAP Cloud Application Programming Model

7

Lesson 3

Introducing the SAP BTP Developer's Guide

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UNIT OBJECTIVES

- Identify the need for side-by-side extensibility
- Describe the SAP Cloud Application Programming Model
- Identify the purpose of the SAP Developer's Guide

Identifying the Need for Side-By-Side Extensibility



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify the need for side-by-side extensibility

Introduction

You are a developer at a company that is using SAP S/4HANA Cloud. Your company has a requirement to extend the standard SAP S/4HANA Cloud solution with custom business logic and UIs.

You want to explore the options for side-by-side extensibility with SAP BTP and SAP S/4HANA Cloud. When you are building extensions in an SAP system, especially in a cloud environment like SAP S/4HANA Cloud, the traditional approach to develop and run custom business logic directly on the application server is no longer possible.

Historically, SAP ERP solutions were extended on-premises directly by adding custom codes to the ABAP application server. The capabilities and paradigms of extending SAP solutions have changed through the shift into a cloud environment. Nevertheless, the standard SAP S/4HANA Cloud system can be adapted to custom business needs.

Various possibilities exist to customize the standard system, like using key-user extensibility - for example, to add custom fields to an SAP Fiori User Interface (UI). However, this only meets the requirements for simple customizations. For more complex requirements, like adding custom business logic, a side-by-side extensibility approach is required. Let's explore this in more detail.

Side-by-Side Extensibility

As introduced previously, the SAP BTP is a cloud-based development platform that focuses on business-centricity. The SAP BTP provides a future-proof cloud(-native) development environment, harmonizing the development experience across SAP products and providing a seamless experience in the cloud.

Three application runtimes, namely the SAP BTP, ABAP Runtime, SAP BTP, Cloud Foundry Runtime, and the SAP BTP, Kyma runtime, provide various options for both ABAP and non ABAP developers on the SAP BTP to develop and run applications in a pro code way. Similarly, for citizen developers, the SAP BTP offers a low-code/no-code (LCNC) suite of products to build enterprise-grade extensions in a fully low-code way.

SAP Build consists of three deeply integrated products, SAP Build Apps, SAP Build Process Automation, and SAP Build Work Zone. It is worth mentioning here is that both sides of the spectrum, pro code and low-code, can be combined to build extensions in a hybrid approach, where pro code developers and citizen developers collaborate in so-called "Fusion Teams" to build extensions.

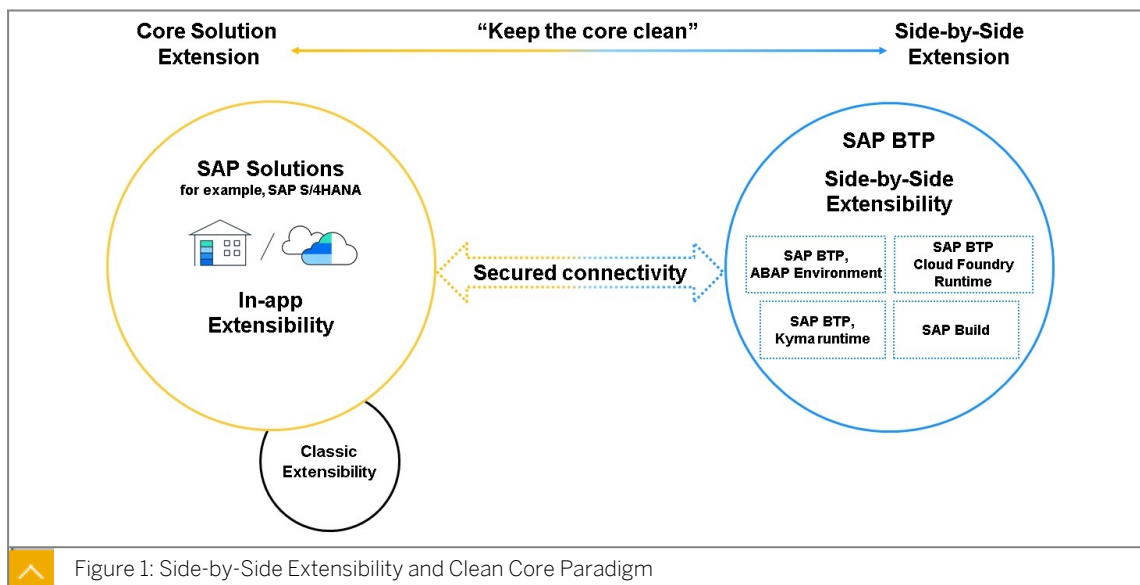


Figure 1: Side-by-Side Extensibility and Clean Core Paradigm

While it is still possible to develop the core solution, it's a proven best practice to "keep the core clean", and rather consume Application Programming Interfaces (APIs) from these systems, and integrate them into your extension applications. This is exactly where the SAP BTP comes into place.

As the cloud-based development platform, the SAP BTP offers a wide range of business-centric capabilities to extend business processes in SAP landscapes and beyond. The foundational plane of the SAP BTP provides developers with various development tools, services, and infrastructure components to build and operate secure, scalable, and reliable cloud-native applications.

Developers have access to development tools, services, and infrastructure components such as the SAP Business Application Studio or the SAP BTP, Kyma runtime.

Extensibility Options in SAP BTP

With the three runtimes on the SAP BTP, different extensibility options are available. For ABAP-based extensions, the SAP BTP, ABAP Environment is the only choice to develop and run the applications. On the other side, building non ABAP-based extensions allows you to use various technology stacks and pure cloud-native development and deployment scenarios.

Programming languages such as (but not limited to) JavaScript (Node.js), Java, TypeScript, Go, Python, and many more can be used to build a custom enterprise-grade extension on the SAP BTP. These applications can be developed either locally in your IDE of your choice, or in the SAP Business Application Studio on the SAP BTP. The deployment can either take part to the SAP BTP, Cloud Foundry Runtime or the SAP BTP, Kyma runtime.



Categorizing the skill set of SAP Cloud Developers

SAP Cloud Developers can specialize in **either ABAP or Non ABAP** development.
Each with its unique set of tools and methodologies.

ABAP-based

- Uses ABAP RESTful Programming Model (RAP)
- Supports only ABAP (Cloud)
- Focuses on S/4HANA extensions
- Development only to On-Prem / ABAP Environment on SAP BTP

Non ABAP-based

- Uses Cloud Application Programming Model (CAP) or SAP Cloud SDK
- Supports JavaScript, TypeScript, Java
- Target multimicroservice-based SaaS and SAP S/4HANA extensions
- Development to Cloud Foundry / Kyma Runtime



Figure 2: ABAP and non-ABAP-based extensions on SAP BTP

In the context of this learning journey, you will only learn about non ABAP-based extensions. More precisely, with the **SAP Cloud Application Programming Model (CAP)** with Node.js (JavaScript) and Cloud Foundry.



LESSON SUMMARY

You should now be able to:

- Identify the need for side-by-side extensibility

Exploring the SAP Cloud Application Programming Model



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the SAP Cloud Application Programming Model

Introduction

Imagine you are a project lead at a rapidly growing company. You face the challenge of digitally transforming the organization, all while needing to integrate with existing SAP solutions seamlessly. This is where the SAP Cloud Application Programming Model (CAP) comes in as your go-to framework.

What Is the SAP Cloud Application Programming Model?

The SAP Cloud Application Programming Model, often abbreviated as CAP, isn't just another tool in your development arsenal. It is a comprehensive framework designed to alleviate most pains traditionally associated with enterprise-grade application development. It comes with a rich set of languages, libraries, and tools for building enterprise-grade services and applications. It guides developers along a 'golden path' of proven [best practices](#) and a great wealth of [out-of-the-box solutions](#) to recurring tasks.

As you delve into the world of CAP, you encounter several key components that hold the architecture together.

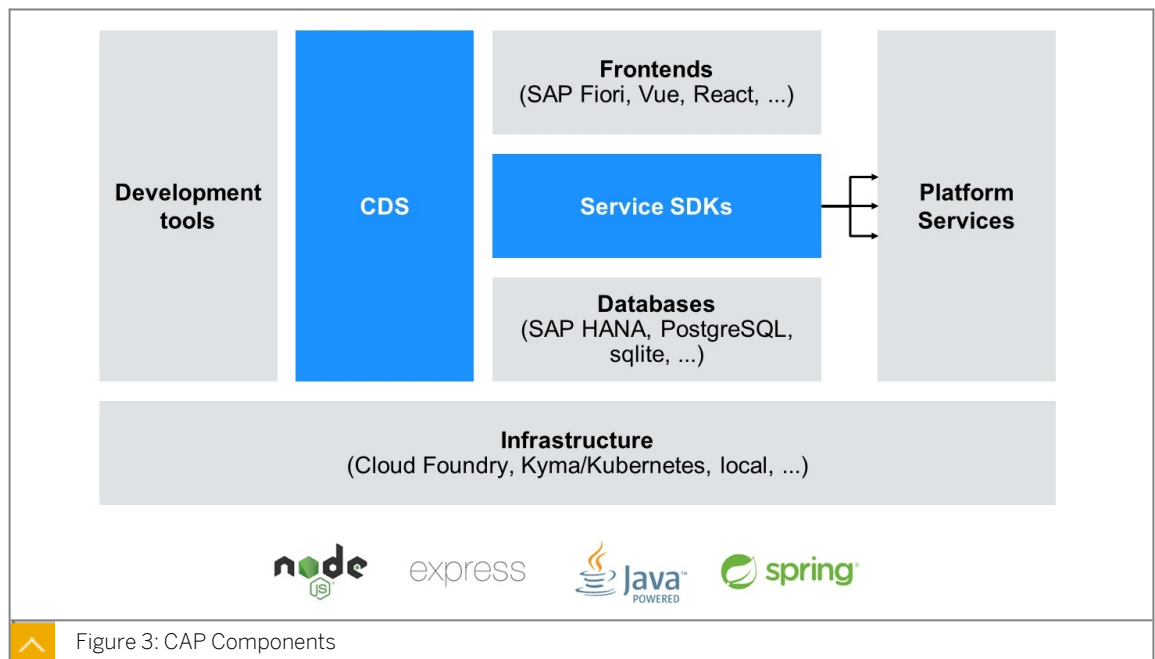


Figure 3: CAP Components

Core Data Services (CDS)

CDS is the data modeling infrastructure, also known as the backbone of the Programming Model, that provides you with the means to capture service definitions and data models. For service definitions and data models, the [Definition Language \(CDL\)](#) is used. More on this in the [Language Reference Documentation](#) in the official CAP documentation (cap.cloud.sap).

Service SDKs

SAP Cloud Application Programming Model has libraries available for both Java and Node.js, which you can use to provide and consume services through synchronous and asynchronous APIs.

The SDKs include out-of-the-box integration to lower-level platform services, such as authentication and credential-flows or on and off-boarding of SaaS tenants.

Development Tools

SAP provides a holistic web-based development environment called the SAP Business Application Studio, which is equipped and preconfigured with all required tools for CAP development. Besides that, local development in VS Code or Java-specific-IDEs is also supported.

Databases

For the persistency layer of your CAP application, you can choose between several available database options. SAP HANA Cloud is the go-to choice. However, other technologies such as PostgreSQL or SQLite (for local development) are possible.

Frontends

As a full stack framework, CAP is open to a variety of frontend technologies. SAP Fiori is fully supported, but other UI frameworks such as Vue.js, Angular, or React can be put on top of your service.

Infrastructure

Two stacks are available: Java and Node.js. Deployments are possible to SAP BTP, Cloud Foundry Runtime, or SAP BTP, Kyma runtime.

Platform Services

SAP BTP service can be integrated and consumed through SDKs.

Overview of Important CAP Resources

As a CAP developer, you should be familiar with some key CAP resources to either get started or to explore advanced scenarios. The first is the [official CAP documentation](#) called **CAPire**.

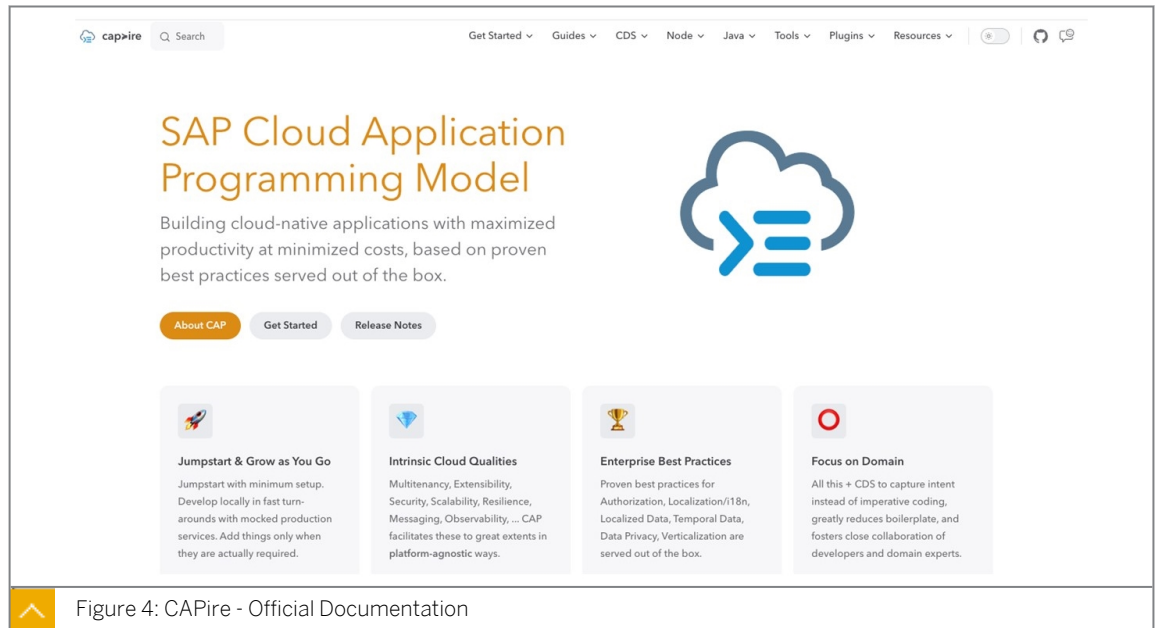


Figure 4: CAPire - Official Documentation

Inside **CAPire**, you find all aspects of CAP for both Node.js and Java. Plenty of beginner and advanced sample projects are available on [GitHub](#).

CAP Node.js Release Cycle

CAP Node.js releases are linked to [Node.js Release Schedules](#). CAP's versioning is also based on [semantic versioning \(semver\)](#), which in turn is based on the version number MAJOR.MINOR.PATCH. The single elements are being incremented as follows:

- **MAJOR**

New **major versions** of CAP are released **every 12 months** (usually around May). When starting a new project, it's highly recommended to adopt the latest major release. Former releases will only receive critical bug fixes only (PATCHES).

- **MINOR**

Active major versions receive **monthly minor updates**. These minor updates contain feature updates that are backwards compatible inside the same major release. So, there are no breaking changes to public APIs of the CAP framework.

- **PATCH**

Between official active and maintenance releases, patch versions are shipped regularly to close bugs.

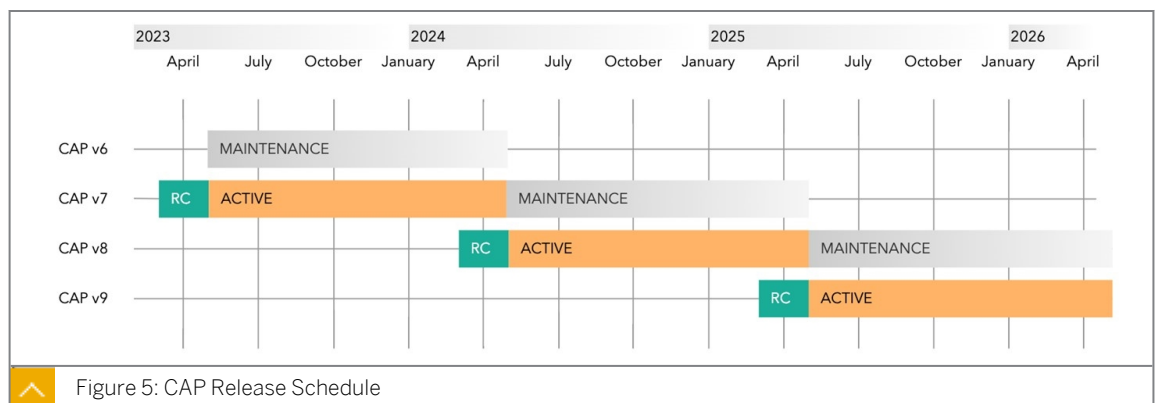


Figure 5: CAP Release Schedule

Read more about the schedule and review change logs in the [release notes](#) section of the CAP documentation.



LESSON SUMMARY

You should now be able to:

- Describe the SAP Cloud Application Programming Model

Introducing the SAP BTP Developer's Guide



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify the purpose of the SAP Developer's Guide

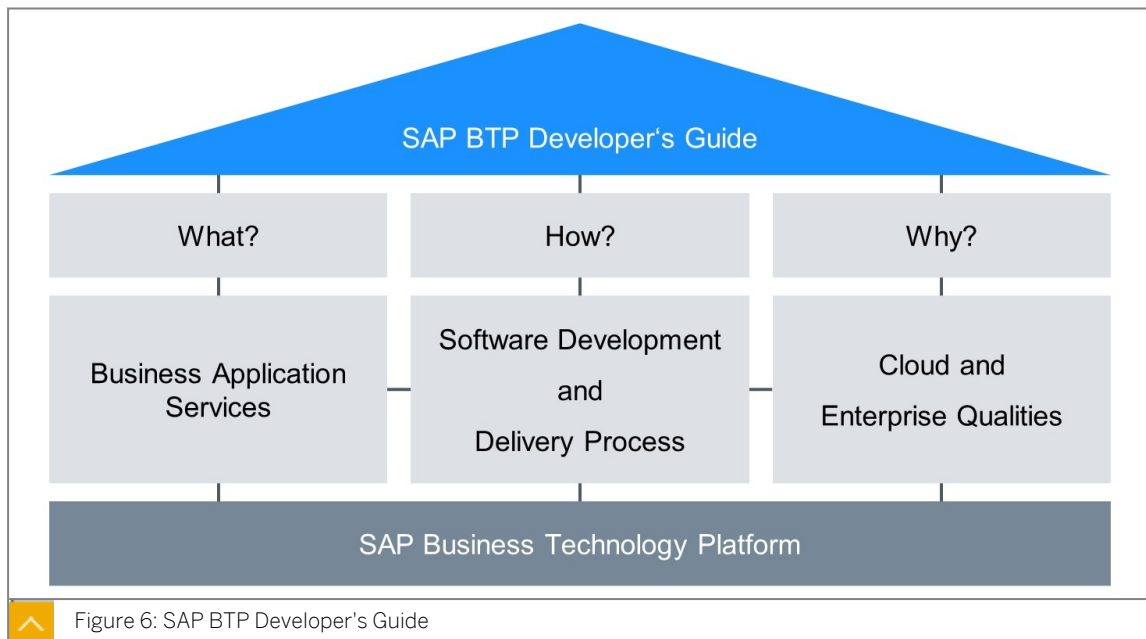
Introduction to the SAP BTP Developer's Guide

Introduction

Feeling overwhelmed by the various options for developing Business Applications with SAP BTP? That's where the SAP Developer's Guide steps in. This guide serves as your crucial companion for crafting business applications on the SAP Business Technology Platform (SAP BTP). Whether you are new to SAP or looking to deepen your understanding, this guide helps you navigate the key components and best practices for successful application development.

Why Use the SAP Developer's Guide?

- **Blueprint for Success:** The guide provides a curated list of services that form the core of SAP business applications on SAP BTP, helping you start with a strong foundation.
- **Automated Setup:** It offers tools for automatic provisioning, management, and configuration of SAP BTP resources, saving you time and effort.
- **Seamless Integration:** Standardized technologies ensure smooth integration across different components, facilitating a cohesive development process.
- **Cloud and Enterprise Quality:** You'll get guidance on delivering robust SAP cloud qualities and intelligent enterprise suite qualities, ensuring your applications are both powerful and reliable.



Key Components

- **SAP Cloud Application Programming Model (CAP):** CAP is a versatile framework supporting Java, JavaScript, and TypeScript, guiding you with built-in best practices and tools for accelerated development. CAP projects focus on the business domain, safeguarding your investments in a rapidly evolving cloud landscape. In this Learning Journey, the focus is on the development with CAP.
- **ABAP Cloud:** This modern approach to ABAP development enables you to create life cycle-stable and cloud-ready business applications. ABAP Cloud promotes new technologies and simplifies upgrade cycles by separating custom code from SAP code.
- **SAP Build Code:** An AI-powered development environment tailored for CAP and SAPUI5 developers. It integrates coding, testing, and application management, streamlining your development process and ensuring adherence to best practices.

Summary

The SAP Developer's Guide is an essential resource for creating business applications on the SAP Business Technology Platform (SAP BTP). It provides a comprehensive roadmap for developers, consolidating best practices, technologies, and methodologies to ensure efficient, scalable, and robust application development.

Further Readings

- [What Is SAP BTP Developer's Guide?](#)
- [Tutorials for SAP Cloud Application Programming Model](#)



LESSON SUMMARY

You should now be able to:

- Identify the purpose of the SAP Developer's Guide

Learning Assessment

1. What is the recommended approach for adding custom business logic to an SAP S/4HANA Cloud system?

Choose the correct answer.

- ☐ A Directly running custom logic on the application server
- ☐ B Using side-by-side extensibility with SAP BTP
- ☐ C Modifying core SAP code
- ☐ D Adding custom code to the ABAP application server

2. What is the programming model used in non ABAP-based extensions on SAP BTP?

Choose the correct answer.

- ☐ A ABAP RESTful Programming Model
- ☐ B SAP Cloud Application Programming Model
- ☐ C SAP Fiori Programming Model

3. What is the primary purpose of the SAP Developer's Guide?

Choose the correct answer.

- ☐ A To provide detailed financial analysis tools for SAP applications.
- ☐ B To streamline the development process for business applications on SAP BTP.
- ☐ C To offer marketing strategies for SAP products.
- ☐ D To teach users how to use SAP ERP systems.

4. Which of the following is NOT a core component of the SAP BTP Developer's Guide?

Choose the correct answer.

- ☐ A ABAP Cloud
- ☐ B SAP Cloud Application Programming Model (CAP)
- ☐ C SAP Build Code
- ☐ D SAP Financial Reporting Tools

Learning Assessment - Answers

1. What is the recommended approach for adding custom business logic to an SAP S/4HANA Cloud system?

Choose the correct answer.

- ☐ A Directly running custom logic on the application server
- ☒ B Using side-by-side extensibility with SAP BTP
- ☐ C Modifying core SAP code
- ☐ D Adding custom code to the ABAP application server

Correct. Using side-by-side extensibility with SAP BTP is the recommended approach for adding custom business logic to an SAP S/4HANA Cloud system.

2. What is the programming model used in non ABAP-based extensions on SAP BTP?

Choose the correct answer.

- ☐ A ABAP RESTful Programming Model
- ☒ B SAP Cloud Application Programming Model
- ☐ C SAP Fiori Programming Model

Correct. The SAP Cloud Application Programming Model is used in non ABAP-based extensions on SAP BTP.

3. What is the primary purpose of the SAP Developer's Guide?

Choose the correct answer.

- ☐ A To provide detailed financial analysis tools for SAP applications.
- ☒ B To streamline the development process for business applications on SAP BTP.
- ☐ C To offer marketing strategies for SAP products.
- ☐ D To teach users how to use SAP ERP systems.

Correct. The primary purpose of the SAP Developer's Guide is to streamline the development process for business applications on SAP BTP.

4. Which of the following is NOT a core component of the SAP BTP Developer's Guide?

Choose the correct answer.

- ☐ A ABAP Cloud
- ☐ B SAP Cloud Application Programming Model (CAP)
- ☐ C SAP Build Code
- ☒ D SAP Financial Reporting Tools

Correct. SAP Financial Reporting Tools is not a core component of the SAP BTP Developer's Guide.

UNIT 2

Setting Up the CAP Project

Lesson 1

Introducing the OData Protocol

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Lesson 2

Explaining JSON/YAML

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Lesson 3

Discovering the End-to-End Use Case

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Lesson 4

Creating CAP-Based Services

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UNIT OBJECTIVES

- Describe the OData standard for web-based applications
- Describe JSON and YAML
- Perform the setup necessary to build your extension project
- Create a CAP-based service

Introducing the OData Protocol



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the OData standard for web-based applications

What Is OData?

Open Data Protocol (OData) is an OASIS standard that defines the best practice for building and consuming RESTful APIs. OData helps you to focus on your business logic, while building RESTful APIs, without having to worry about the approaches to define request and response headers, status codes, HTTP methods, URL conventions, media types, payload formats, query options, and so on. It is an open standard that is defined by the OASIS consortium.

OData also guides you in tracking changes, defining functions or actions for reusable procedures, sending asynchronous or batch requests, and so on. Also, OData provides a facility for extensions to fulfill any custom needs of your RESTful APIs.

OData RESTful APIs are easy to consume. The OData metadata, a machine-readable description of the data model of the APIs, enables the creation of powerful generic client proxies and tools. Some of these can help you interact with OData, even without knowing anything about the protocol.

OData Basics

One of the main features of OData is that it uses the existing HTTP verbs GET, PUT, POST, and DELETE against addressable resources identified in the URI. Conceptually, OData is a way of performing database-style create, read, update, and delete operations on resources by using HTTP verbs:



- **GET:** Get the resource (a collection of entities, a single entity, a structural property, and so on).
- **POST:** Create a new resource.
- **PUT:** Update an existing resource by replacing it with a complete instance.
- **PATCH:** Update an existing resource by replacing part of its properties with a partial instance.
- **DELETE:** Remove the resource.

OData Service

An **OData service** is a logical data model; it describes entities (resources) using associations and operations. The most important point is that the OData service forms a kind of contract between the UI and the back-end system side, helping to bring together developers on both sides.

OData currently supports two formats for representing the resources that it exposes:

- The XML-based AtomPub
- The JSON formats

JSON has less protocol overhead than the Atom Publishing protocol. JSON can easily be consumed with JavaScript and by SAPUI5.

Each OData service is represented by a URI (Uniform Resource Identifier), called the service root URI. A URI is a uniform resource identifier, which is a string of characters used to identify a resource. More precisely, each resource can be accessed using a URL (Uniform Resource Locator). A uniform resource locator describes how to access the resource. This type of identification enables the interaction with representations of the resource across a network using specific protocols like OData.

Types of Documents Associated with Each OData Service

There are two types of documents associated with each OData service:



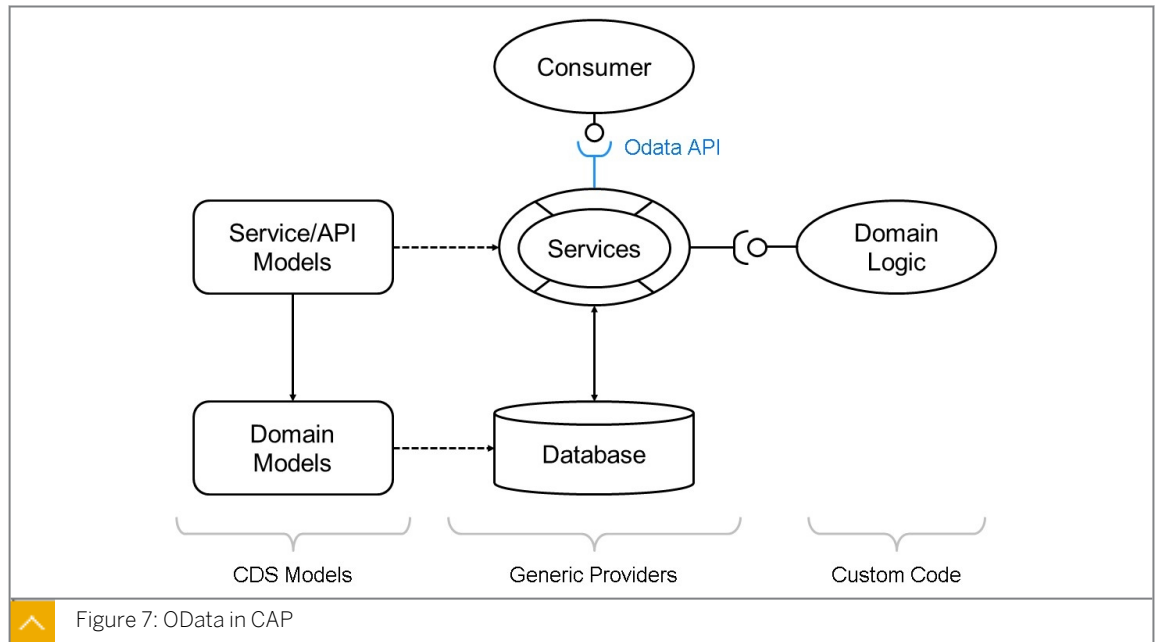
- The service document
- The service metadata document

The service document lists entity sets, functions, and singletons that can be retrieved. Clients can use the service document to navigate the model in a hypermedia-driven fashion. The service document is available at `http://<host>:<port>/<service>/`.

The metadata document describes the types, sets, functions, and actions understood by the OData service. Clients can use the metadata document to understand how to query and interact with entities in the service. The service metadata document is available at `http://<host>:<port>/<service>/$metadata`. The URL returns XML metadata of the service (Entity data model). The response of a service metadata document only supports XML.

OData and CAP

In a CAP-based application, services are articulated through Core Data Services (CDS) models and are managed by the CAP runtime environment. In the CAP framework, every functional component is viewed as a service. These services represent the interactive characteristics of a domain, including the entities they expose, the actions they enable, and the events they trigger.

**Note:**

While the default protocol for exposed services in CAP is OData, you can also use [Protocol Adapters](#) in CAP to overwrite the default behavior. Protocols such as **REST** and **GraphQL** are supported out-of-the-box. Also, it's also possible to define your own **Custom Protocol Adapters**.

Summary

You now have a more profound understanding of the OData protocol and you can describe its key concepts. You have also learned about OData in CAP and have heard about the possibility to use different protocols through protocol adapters.

Further Reading

- [OData - the Best Way to REST](#)
- [Understand OData in 6 steps](#)
- [Providing Services in CAP](#)
- [Protocol Adapters](#)

**LESSON SUMMARY**

You should now be able to:

- Describe the OData standard for web-based applications

Explaining JSON/YAML



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe JSON and YAML

JSON and YAML

Business scenario

You find yourself needing to create configuration files for various cloud-native services. Should you use JSON or YAML? This lesson guides you through the specifics of each, helping you make an informed decision.

What Are JSON and YAML?

JSON Overview

[JSON](#) (JavaScript Object Notation) is an open-standard format for data storage and exchange. It employs easy-to-read text to encapsulate data objects, which consist of key-value pairs and arrays. Frequently used for web applications to communicate with servers, JSON serves as a versatile data interchange medium.

Though influenced by JavaScript's syntax, JSON operates independently and enjoys support across various programming languages. Files using this format typically have a `.json` extension.

JSON is fundamentally built upon two key structures:

- A set of **key-value pairs**: This is often implemented as an 'object' in multiple programming languages.
- A sequentially **ordered series of values**: Most languages interpret this as an 'array'.

The following figure, JSON Representation Describing a Business Partner, shows a possible JSON representation describing a Business Partner.



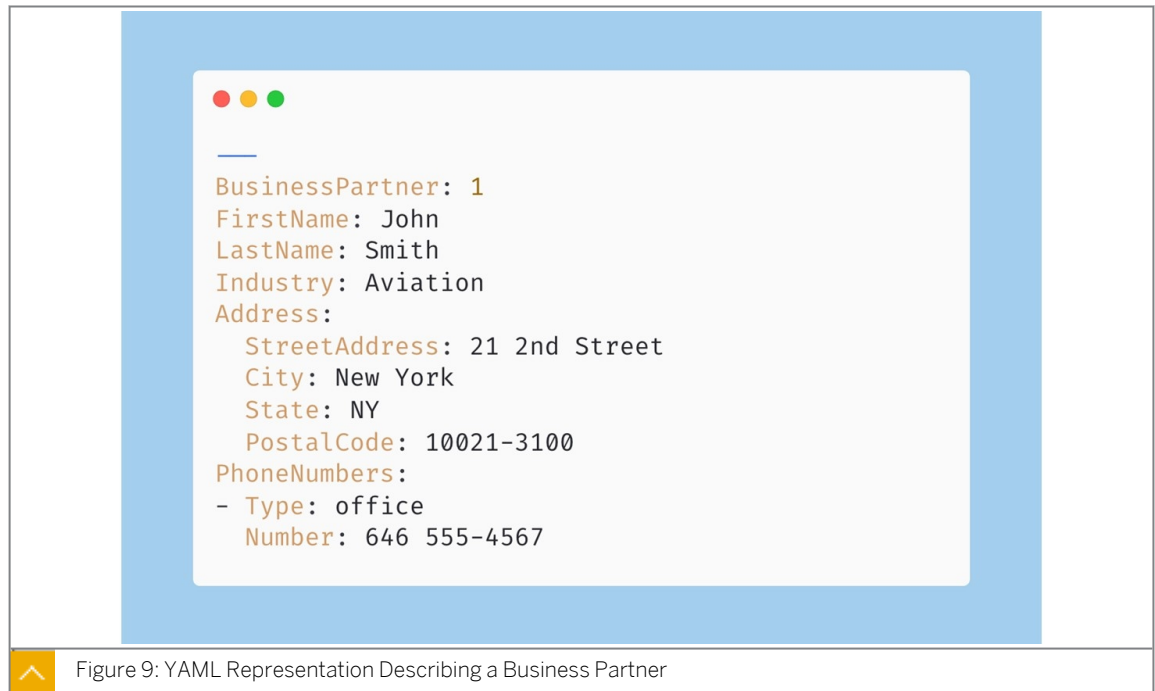
Figure 8: JSON Representation Describing a Business Partner

YAML Overview

YAML is a human-friendly, cross-language, Unicode-based data serialization language designed around the common native data types of agile programming languages. It is broadly useful for programming needs ranging from configuration files to internet messaging to object persistence to data auditing. YAML was specifically created to work well for common use cases such as; the following: configuration files, log files, interprocess messaging, cross-language data sharing, object persistence, and debugging of complex data structures. When data is easy to view and understand, programming becomes a simpler task. YAML filenames use the extension `.yaml` or `.yml`.

YAML is a strict JSON superset and includes extra features such as the notion of tagging data types, support for non hierarchical data structures, the option to structure data with indentation, and multiple forms of scalar data quoting. YAML is an open format.

The following figure, *YAML Representation Describing a Business Partner*, shows a possible YAML representation describing a Business Partner.



YAML in Relation to JSON

Both JSON and YAML aim to be human-readable data interchange formats. However, JSON and YAML have different priorities.

JSON's foremost design goal is simplicity and universality. Therefore, JSON is trivial to generate and parse, at the cost of reduced human readability. It also uses a lowest common denominator information model, ensuring any JSON data can be easily processed by every modern programming environment.

In contrast, YAML's foremost design goals are human readability and support for serializing arbitrary native data structures. Therefore, YAML allows for readable files, but is more complex to generate and parse. In addition, YAML ventures beyond the lowest common denominator data types, requiring more complex processing when crossing between different programming environments.

YAML can therefore be viewed as a natural superset of JSON, offering improved human readability, and a more complete information model. It is also the case in practice; every JSON file is also a valid YAML file. This makes it easy to migrate from JSON to YAML if or when the additional features are required.

Summary

You now have a more profound understanding of JSON and YAML.

Further Reading

- [JSON Specification](#)
- [YAML Specification](#)



LESSON SUMMARY

You should now be able to:

- Describe JSON and YAML

Discovering the End-to-End Use Case



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Perform the setup necessary to build your extension project

Introduction

Business Case

Let's begin by considering the following scenario.

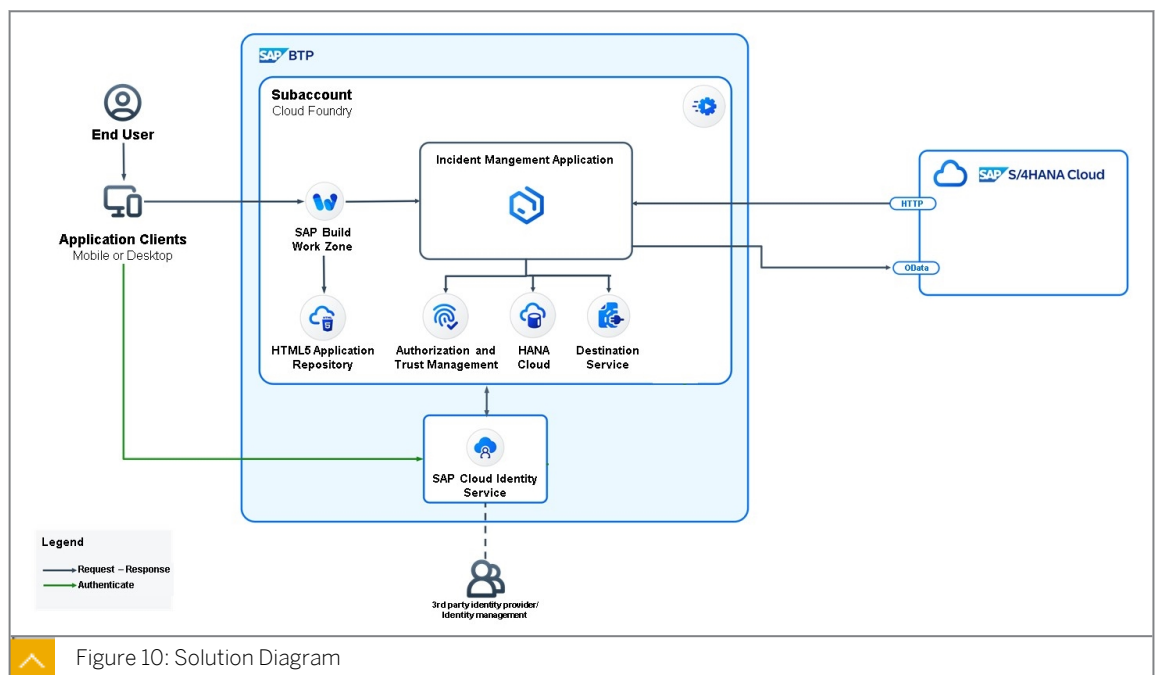


Figure 10: Solution Diagram

Let's first take a closer look at the underlying business case. Your company ACME is an electronics company, that hires call center support representatives to process and manage customer incidents.

A call center support representative (Processor) receives a phone call from an existing customer and creates a new incident in an incident management application on behalf of the customer. The newly created incident is based on a customer complaint received during the phone call so the call center support representative also adds the conversation with the customer to the incident.

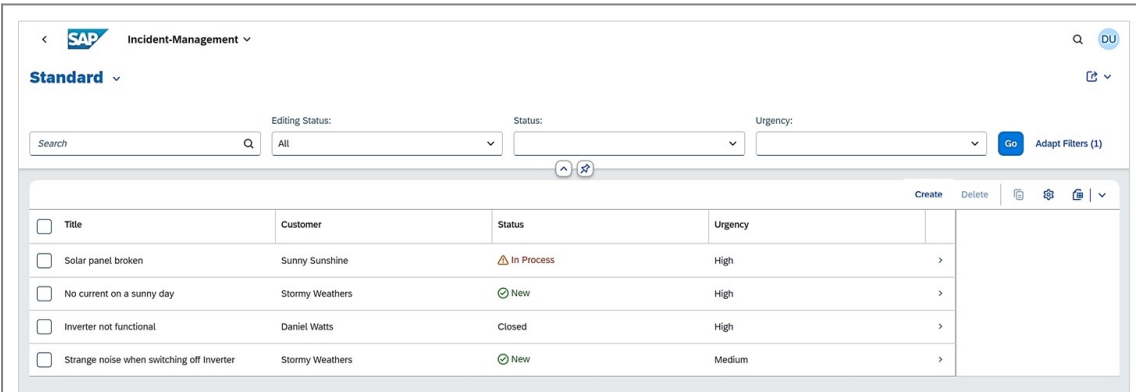
This application enables the interaction between the members of the support team who are working on the customer's incident. The application is designed to allow support team

members to initiate incident reports, which are then processed by other support team members.

Preview of the Final Application

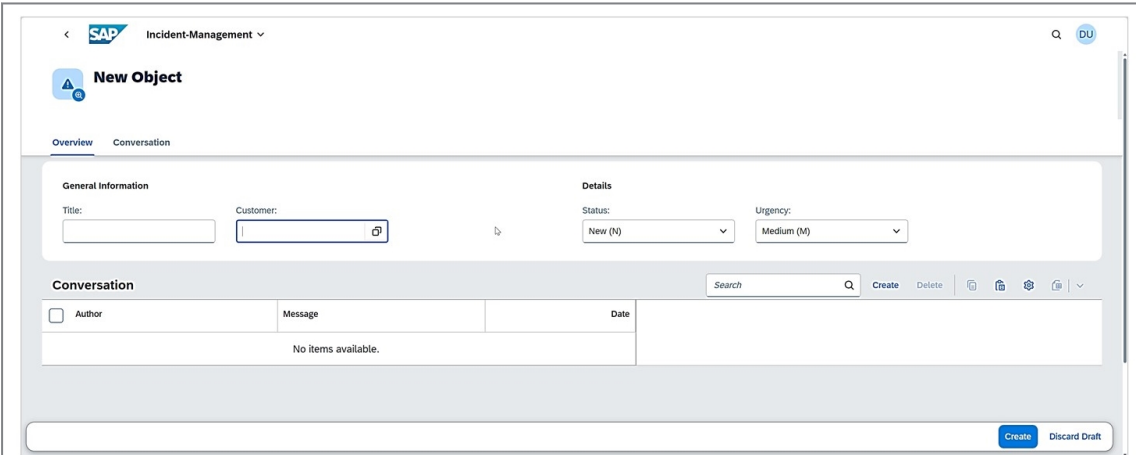
The frontend is built with SAP Fiori Elements and consists of two pages: A List Page, listing all incidents, and an Object Page for displaying further information for each incident entry.

This is the application's homepage. It displays current incidents needing the incident manager's attention. It also indicates the affected customer and the incident's status and urgency. Users can create, edit, and delete incidents here.

Title	Customer	Status	Urgency
Solar panel broken	Sunny Sunshine	In Process	High
No current on a sunny day	Stormy Weathers	New	High
Inverter not functional	Daniel Watts	Closed	High
Strange noise when switching off inverter	Stormy Weathers	New	Medium

Figure 11: List Page

New Object

Overview Conversation

General Information

Title:

Customer:

Details

Status:

Urgency:

Conversation

Author	Message	Date
No items available.		

Create Discard Draft

Figure 12: Object Page

Users can create new incidents and input information on the object page. The system fetches customer data from SAP S/4HANA Cloud.

To view or edit an incident, select it to open the details page.

What's Next?

The following units guide you through an end-to-end development process from development to deployment.



LESSON SUMMARY

You should now be able to:

- Perform the setup necessary to build your extension project

Creating CAP-Based Services



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Create a CAP-based service



LESSON SUMMARY

You should now be able to:

- Create a CAP-based service

Learning Assessment

1. What is OData?

Choose the correct answer.

- ☐ A A standard to access data through RESTful APIs
- ☐ B A standard to create user interfaces (UI) for applications
- ☐ C A standard to share data under a creative commons license

2. Which of the following does OData use to address and access data feed resources?

Choose the correct answer.

- ☐ A URL
- ☐ B URN
- ☐ C URI

3. Which document types are associated with an OData service?

Choose the correct answers.

- ☐ A Service manifest document
- ☐ B Service document
- ☐ C Service description document
- ☐ D Service metadata document

4. JSON is based on which programming language?

Choose the correct answer.

- ☐ A Java
- ☐ B Julia
- ☐ C JCL
- ☐ D JavaScript

5. What are the two structures JSON is built on?

Choose the correct answers.

- ☐ A Collections of name/value pairs
- ☐ B Collections of value/value pairs
- ☐ C Unordered list of strings
- ☐ D Ordered list of values

6. What is the relationship between YAML and JSON?

Choose the correct answer.

- ☐ A YAML and JSON are unrelated.
- ☐ B JSON is a superset of YAML.
- ☐ C YAML is a superset of JSON.

7. Which of the following supports non hierarchical data?

Choose the correct answer.

- ☐ A YAML
- ☐ B JSON
- ☐ C Neither YAML nor JSON

8. Which of the following statements are correct?

Choose the correct answers.

- ☐ A Each JSON file is a valid YAML file.
- ☐ B Each YAML file is a valid JSON file.
- ☐ C JSON's foremost design goal is support for serializing arbitrary native data structures.
- ☐ D YAML's foremost design goal is support for serializing arbitrary native data structures.

9. What is the default IDE for SAP's multicloud environment?

Choose the correct answer.

- ☐ A Eclipse
- ☐ B Oxygen
- ☐ C SAP Business Application Studio
- ☐ D Microsoft Visual Code

10. Which dev space type should you select to extend SAP S/4HANA with a CAP project?

Choose the correct answer.

- ☐ A SAP Fiori
- ☐ B SAP HANA Native Application
- ☐ C Full Stack Cloud Application

11. Which command do you use to install dependencies in your project?

Choose the correct answer.

- ☐ A cds
- ☐ B yum
- ☐ C bash
- ☐ D npm

12. In data models, are namespaces optional or mandatory?

Choose the correct answer.

- ☐ A Mandatory
- ☐ B Optional

13. What is the difference between entities and types?

Choose the correct answer.

- ☐ A Types represent data elements, entities describe aspects of types.
- ☐ B Entities represent data, types describe properties of entity elements.

Learning Assessment - Answers

1. What is OData?

Choose the correct answer.

- ☒ A A standard to access data through RESTful APIs
- ☐ B A standard to create user interfaces (UI) for applications
- ☐ C A standard to share data under a creative commons license

Correct. OData is a standard to access data through RESTful APIs.

2. Which of the following does OData use to address and access data feed resources?

Choose the correct answer.

- ☐ A URL
- ☐ B URN
- ☒ C URI

Correct. OData uses URI to address and access data feed resources.

3. Which document types are associated with an OData service?

Choose the correct answers.

- ☐ A Service manifest document
- ☒ B Service document
- ☐ C Service description document
- ☒ D Service metadata document

Correct. The following document types are associated with an OData service: "Service document" and "Service metadata document".

4. JSON is based on which programming language?

Choose the correct answer.

- ☐ A Java
- ☐ B Julia
- ☐ C JCL
- ☒ D JavaScript

Correct. JSON is based on JavaScript.

5. What are the two structures JSON is built on?

Choose the correct answers.

- ☒ A Collections of name/value pairs
- ☐ B Collections of value/value pairs
- ☐ C Unordered list of strings
- ☒ D Ordered list of values

Correct. The two structures JSON is built on are as follows: Collections of name/value pairs, and an ordered list of values.

6. What is the relationship between YAML and JSON?

Choose the correct answer.

- ☐ A YAML and JSON are unrelated.
- ☐ B JSON is a superset of YAML.
- ☒ C YAML is a superset of JSON.

Correct. The relationship between YAML and JSON is this: YAML is a superset of JSON.

7. Which of the following supports non hierarchical data?

Choose the correct answer.

- ☒ A YAML
- ☐ B JSON
- ☐ C Neither YAML nor JSON

Correct. YAML supports non hierarchical data.

8. Which of the following statements are correct?

Choose the correct answers.

- ☒ A Each JSON file is a valid YAML file.
- ☐ B Each YAML file is a valid JSON file.
- ☐ C JSON's foremost design goal is support for serializing arbitrary native data structures.
- ☒ D YAML's foremost design goal is support for serializing arbitrary native data structures.

Correct. The statements: "Each JSON file is a valid YAML file", and "YAML's foremost design goal is support for serializing arbitrary native data structures" are correct.

9. What is the default IDE for SAP's multicloud environment?

Choose the correct answer.

- ☐ A Eclipse
- ☐ B Oxygen
- ☒ C SAP Business Application Studio
- ☐ D Microsoft Visual Code

Correct. The default IDE for SAP's multicloud environment is SAP Business Application Studio.

10. Which dev space type should you select to extend SAP S/4HANA with a CAP project?

Choose the correct answer.

- ☐ A SAP Fiori
- ☐ B SAP HANA Native Application
- ☒ C Full Stack Cloud Application

Correct. You should use the dev space: "Full Stack Cloud Application".

11. Which command do you use to install dependencies in your project?

Choose the correct answer.

- ☐ A cds
- ☐ B yum
- ☐ C bash
- ☒ D npm

Correct. To install dependencies in your project, you use the command `npm`.

12. In data models, are namespaces optional or mandatory?

Choose the correct answer.

- ☐ A Mandatory
- ☒ B Optional

Correct. In data models, namespaces are optional.

13. What is the difference between entities and types?

Choose the correct answer.

- ☐ A Types represent data elements, entities describe aspects of types.
- ☒ B Entities represent data, types describe properties of entity elements.

Correct. Entities represent data, types describe properties of entity elements.

UNIT 3

Serving User Interfaces in CAP

Lesson 1

Serving User Interfaces in CAP

41

Lesson 2

Generating User Interfaces

45

UNIT OBJECTIVES

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori tools
- Generate a User Interface (UI) using SAP Fiori Elements

Serving User Interfaces in CAP



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori tools

Frontend Capabilities

Support for User Interfaces

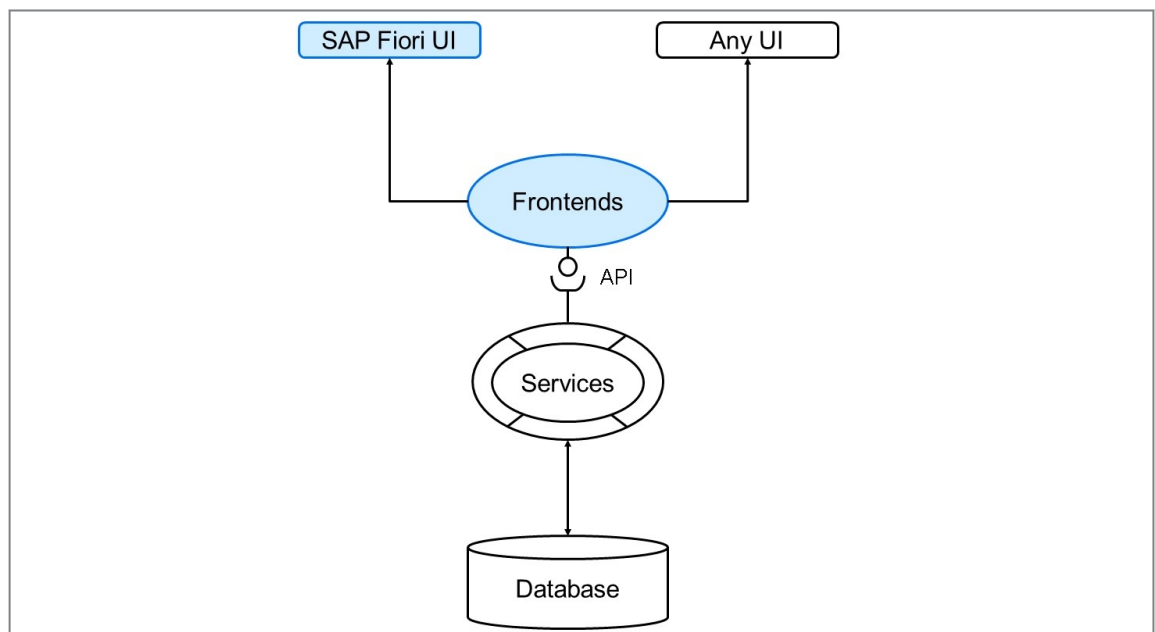


Figure 13: Frontends in CAP

CAP supports the integration of user interfaces. All services provided by CAP cannot only be consumed by other services, but also from user interfaces (UIs). CAP is open and services can be consumed by any UI framework by sending standard AJAX requests. Have a look at the [cap-samples repository](#) for a working [Vue.js example](#).

CAP and SAP Fiori

CAP provides **out-of-the-box support for SAP Fiori elements UIs**, for example, with respect to SAP Fiori annotations and advanced features such as search, value helps, and SAP Fiori draft. Drafts are used to prevent data loss in case an app terminates unexpectedly or to prevent an object being edited by multiple users concurrently.

A single CAP project can contain multiple SAP Fiori apps. Despite the fact that SAP UI5 is also possible, we focus here on **SAP Fiori elements**. Special UI-specific annotations make it possible to enrich the exposed service with instructions for SAP Fiori elements. SAP Fiori elements also read the metadata document and the entity definition to generate the UI. All SAP Fiori apps of a CAP application should go into the `app` folder.

For speeding up the UI development, especially for SAP Fiori Elements, the **SAP Fiori tools** provide advanced support for adding and designing SAP Fiori apps to existing CAP projects. They also come with plenty of **productivity tools**, which encompass visual tools for designing SAP Fiori pages.

SAP Fiori Tools

SAP Fiori Tools are preinstalled in the SAP Business Application Studio and can be used out-of-the-box. You can also install the extension in Visual Studio Code (VS Code), by installing the [SAP Fiori Tools - Extension Pack](#).

Page Map Editor

The Page Map is a feature that comes with the SAP Fiori Tools and can be used as a visual productivity tool to change and design the SAP Fiori User Interfaces. Behind the scenes, it generates all the required annotation files, which can then be checked-in into version control.

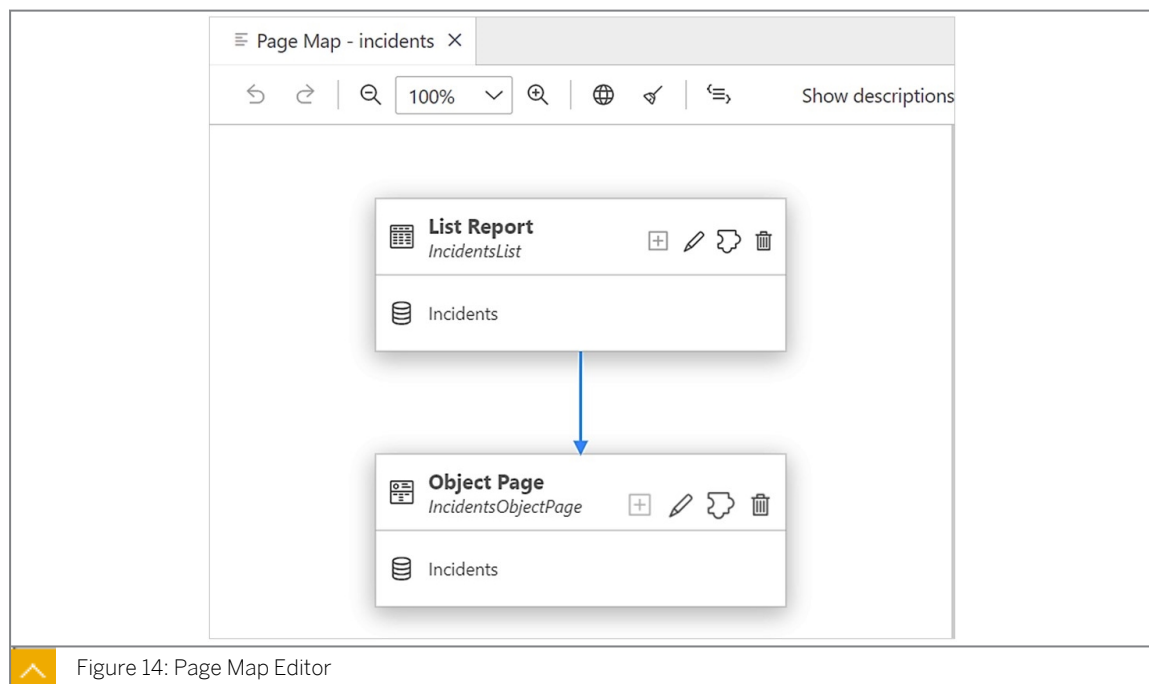


Figure 14: Page Map Editor

The following visualizes the capabilities of the Page Map for an SAP Fiori List Page, where the entire SAP Fiori page, including the table fields and more, can be adjusted. In the image, you see, for example, that the Header can be set dynamically to the respective title value.

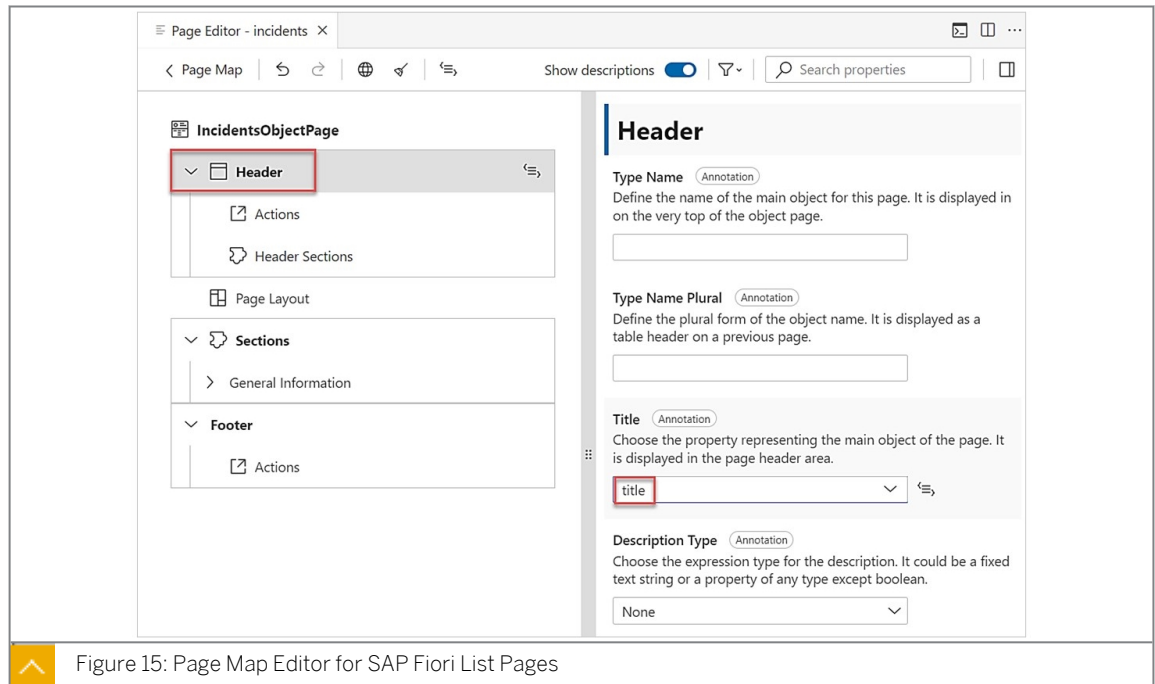


Figure 15: Page Map Editor for SAP Fiori List Pages

Over the course of the next exercises, you will get your hands on these tools.

Summary

You are now familiar with how different UIs can be served in a CAP project and you have learned about the out-of-the-box support for SAP Fiori UIs and the SAP Fiori Tools to speed up your UI development.

Further Reading

- [CAP Docs: Serving Fiori UIs](#)
- [Learning the Basics of SAP Fiori](#)
- [Developing SAP UI5 and SAP Fiori Elements Applications](#)
- [Developing an SAP Fiori Elements App based on a CAP OData v4 Service](#)



LESSON SUMMARY

You should now be able to:

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori tools

Generating User Interfaces



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Generate a User Interface (UI) using SAP Fiori Elements



LESSON SUMMARY

You should now be able to:

- Generate a User Interface (UI) using SAP Fiori Elements

Learning Assessment

1. *Choose the correct answers.*

2. Which statement is true about CAP's interaction with user interfaces?

Choose the correct answer.

- ☐ A CAP services can only be consumed by other back-end services.
- ☐ B CAP services can be consumed by any UI framework through standard AJAX requests.
- ☐ C CAP services are limited to SAP Fiori UIs.

3. What tooling is preinstalled in SAP Business Application Studio to speed up SAP Fiori UI development?

Choose the correct answer.

- ☐ A Vue.js Extensions
- ☐ B SAP Fiori Tools
- ☐ C Bootstrap Toolkit

4. Which of the following is a feature of SAP Fiori Tools designed to visually assist in SAP Fiori UI development?

Choose the correct answer.

- ☐ A AJAX Mapper
- ☐ B Page Map
- ☐ C List Page Designer

5. Which filter fields are set in the exercise: Generate a User Interface Using SAP Fiori Elements to define which properties are exposed as search fields in the header bar above the list?

Choose the correct answer.

- ☐ A urgency_code and filter_code
- ☐ B status_code and filter code
- ☐ C status_code and urgency_code

Learning Assessment - Answers

1. *Choose the correct answers.*

2. Which statement is true about CAP's interaction with user interfaces?

Choose the correct answer.

- ☐ A CAP services can only be consumed by other back-end services.
- ☒ B CAP services can be consumed by any UI framework through standard AJAX requests.
- ☐ C CAP services are limited to SAP Fiori UIs.

Correct. CAP services can be consumed by any UI framework through standard AJAX requests.

3. What tooling is preinstalled in SAP Business Application Studio to speed up SAP Fiori UI development?

Choose the correct answer.

- ☐ A Vue.js Extensions
- ☒ B SAP Fiori Tools
- ☐ C Bootstrap Toolkit

Correct. SAP Fiori Tools are preinstalled.

4. Which of the following is a feature of SAP Fiori Tools designed to visually assist in SAP Fiori UI development?

Choose the correct answer.

- ☐ A AJAX Mapper
- ☒ B Page Map
- ☐ C List Page Designer

Correct. The Page Map is part of SAP Fiori Tools.

5. Which filter fields are set in the exercise: Generate a User Interface Using SAP Fiori Elements to define which properties are exposed as search fields in the header bar above the list?

Choose the correct answer.

- ☐ A urgency_code and filter_code
- ☐ B status_code and filter code
- ☒ C status_code and urgency_code

Correct. You set the filter fields: status_code and urgency_code.

UNIT 4

Adding Custom Business Logic

Lesson 1

Explaining Event Handling in CAP

53

Lesson 2

Explaining the Need for Custom Business Logic

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Lesson 3

Describing Error Handling

59

Lesson 4

Adding Custom Business Logics

63

Lesson 5

Adding Local Launch Pages

65

UNIT OBJECTIVES

- Evaluate when to use event handlers
- Explain the need for custom business logic
- Explain error handling
- Add custom business logic
- Add a local launch page

Explaining Event Handling in CAP



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Evaluate when to use event handlers

Event Handling - CAP Service SDK for Node.js

Usage Scenario

Your company is planning to build an extension application using the SAP Cloud Application Programming Model (CAP). The generic service handlers that the framework provides for standard CRUD operations (CREATE, READ, UPDATE, DELETE) do not fully satisfy the application's requirements. You want to implement custom business logic on top of the standard functionality. For that, you must understand the concept of event handlers in CAP.

In CAP, everything that happens at runtime is an event that is sent to a service. They are a powerful means to extend CAP. An event handler is simply a method that is executed when something happens in the application.

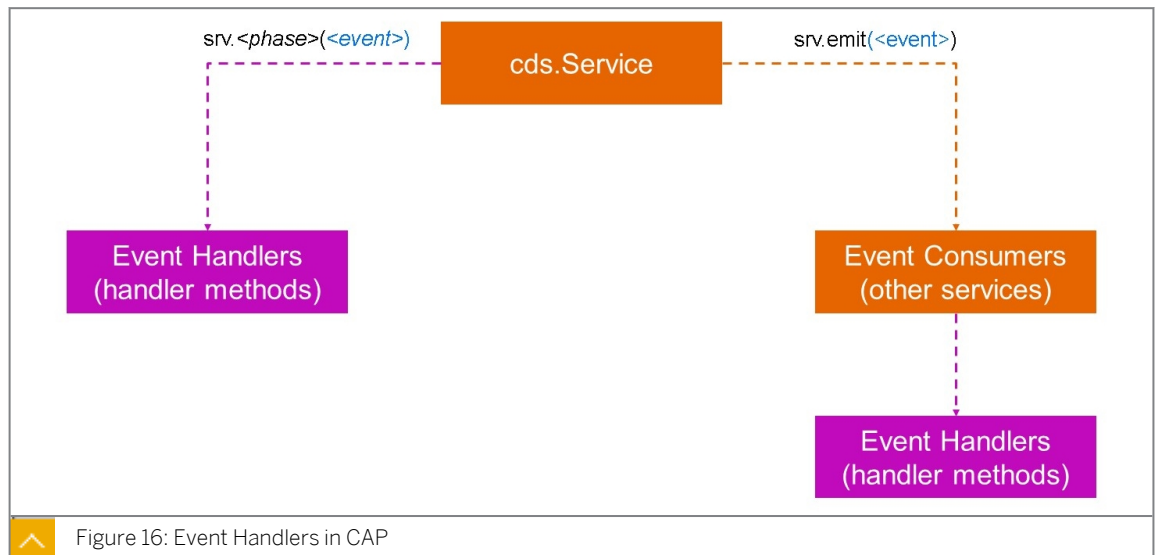


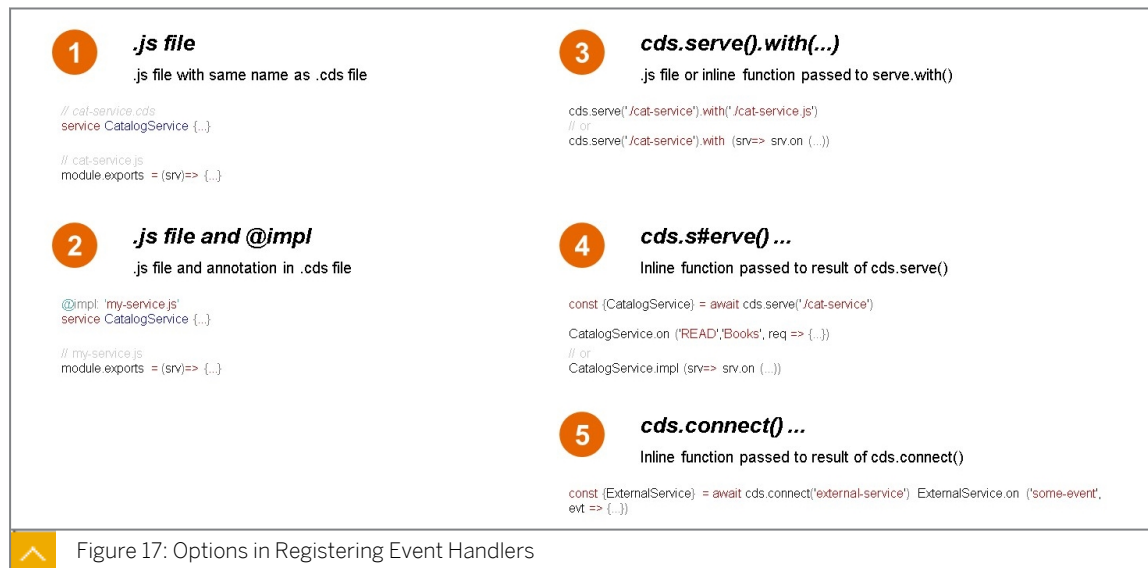
Figure 16: Event Handlers in CAP

If you need a specific service to react to a specific event, you register an event handler using `srv.<phase>(<event>)`, where:

- `srv` is the instance of the service that you are extending,
- `<phase>` is one of `on`, `before`, or `after` (see the following section **Event Phases**) and
- `<event>` is any kind of named event as a string (for example `'READ'`).

When a service has an event handler for a specific event, it becomes a consumer for that event. Using `srv.emit(<event>)`, a service can send arbitrary events. These events then get consumed by other services that have event handlers registered for the respective event.

Introducing Registering Event Handlers



Option 1

In this method, the Javascript file is placed next to the CDS (.cds) file used to define the service, the Javascript file needs to have the same name as the .cds file, this way the framework hooks up the implementation to the service file.

Option 2

Here we set the link through the impl annotation in your CDS model file (.cds), where the respective service implementation can be found. This is useful if you have diverging file names or you want to make it very explicit that the two files belong together.

Option 3 and 4

These are fairly advanced, and used with the CDS serve API to bootstrap your services on your own.

Option 5

This is used when dealing with external services.

Request Objects



 req.method HTTP method of incoming request, e.g. POST, GET, PUT ...	 req.target Refers to the current request's target entity definition, if any	<pre> srv.before(['CREATE', 'UPDATE'], 'Orders', (req) => { const tx = cds.transaction(req) const order = req.data return Promise.all(order.Items.map(each => tx.run(UPDATE(Books).where({ ID: each.book_ID }) .and('stock >=', each.amount) .set('stock -=', each.amount))).then(affectedRows => { if (!affectedRows) { req.error(409, 'insufficient stock') } }) })) req.on('succeeded', () => {...}) // request succeeded req.on('failed', () => {...}) // request failed req.on('done', () => {...}) // request succeeded/failed </pre>
 req.query Captures the incoming request as a CQN query object	 req.data Captures all query parameters as well as http post bodies as a single object	
 req.error Returns an error message to the client. If the first argument is a number, it is used as the HTTP response header code	 req.on Use this method to register handlers, executed when the whole request is finished	
 Event handlers registered via service.on, service.before, service.after all accept a request object as argument, which provides single point of API contact for accessing request input, reading/writing data, and sending responses.		

Figure 18: Request Objects, Overview

Request objects are passed in the event handlers. They provide all sorts of information about the context, like the request data or the method, like get post and so on. Request objects are also used to provide error messages back to the client, or to register another set of handlers for the request life cycle, like when the request has completed, succeeded, or failed.

An Example

Think of a simple application that manages your company's IT inventory. For each asset category, there is an entity - for example, Notebooks, Phones, Tablets. The entities are exposed using an `inventory` service. The service provides an OData API that enables you to interact with the entities. Let's say you want to see the current inventory of notebooks. You perform a GET request to the `inventory/Notebooks` service. Within CAP, a READ event is triggered for the `Notebooks` entity. There is a built-in event handler (also known as *Generic Provider*) that retrieves the requested data from the database on a READ event for the `Notebooks` entity.

Making Use of Event Handlers

CAP handles all CRUD events (`CREATE`, `READ`, `UPDATE`, `DELETE`) out-of-the-box. You do not need to take any further steps after defining your entities and services. However, oftentimes the standard functionality does not fulfill all of your requirements. You want to implement custom logic. In these cases, you use the [handler registration API](#) of the CAP Service SDK for Node.js.

Extending the Example Above

Let's assume you are building a UI on top of your inventory service. It displays when a device is eligible for replacement. Whether a device is eligible for replacement depends on device type, date of acquisition, or country - thus there is no easy answer. Your service must uncover it. Within your entities, there is a Boolean field `eligible_for_replacement`, which is set to `false` by default.

Whenever there is a `READ` event for any of the entities in your `inventory` service, after the entities have been read, you want to have a custom event handler finding out, whether the individual devices are eligible for replacement or not. It could look like the following:

```
cds.service('inventory-service') .with (function() {
  this.after('READ', '*', (devices)=>{
    for (let each of devices) {
      var deviceAge = calculateDeviceAgeYears(each)
      if (deviceAge >= 4) {
        each.eligible_for_replacement = true
      } else {
        each.eligible_for_replacement = false
      }
    }
  })
})
```

The code defines that after each `READ` of any (*) entity in your `inventory` service, the eligibility for replacement is calculated. The method loops through each line of the data that was fetched by the generic service handler. Note that instead of registering the handler method for any entity (*), you could also register the event handler for a specific entity - for example, `Notebooks`.

In this simple example, eligibility for replacement only depends on the age of the device, but you can grasp that almost anything is possible here.

Event Phases

Events are processed in three phases that are executed consecutively: **Before, On, and After**. When registering an event handler, the phase in which the event handler is called must be specified. In the previous example the handler method was specified for the After phase. It is possible to register multiple event handlers for each event phase. Likewise, a single handler can handle multiple events.

In the case of an external service, there wouldn't be anything that the framework could do, so you must provide an 'on' handler, but you can also use the 'on' hook to override generic CRUD handlers. However, it is recommended to use a generic implementation and only diverge from it if you need different logic.

Summary

You can now explain the concept of event handlers in CAP.

Also, you are able to evaluate whether a custom event handler is required, and you are able to perform the basic implementation of custom event handlers.

Further Reading

For more information, consult the following:

- [Event Handling](#)
- [CAP Service SDK for Node.js](#)



LESSON SUMMARY

You should now be able to:

- Evaluate when to use event handlers

Explaining the Need for Custom Business Logic



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the need for custom business logic

The Need for Custom Business Logic

Usage Scenario

While CAP provides a high-level abstraction for building cloud (native/side-by-side) applications, there will always be cases where custom logic is essential to meet the unique needs of your business or to integrate with external systems. Custom logic in CAP allows you to extend and customize the default behaviors of the framework to create a fully tailored and functional application.

Introducing Service API for Custom Logic

Custom Code is the logic that you can add to the application to express things like data validation, enrichment, calculations, integration with external systems, or any other custom processing that your application requires.

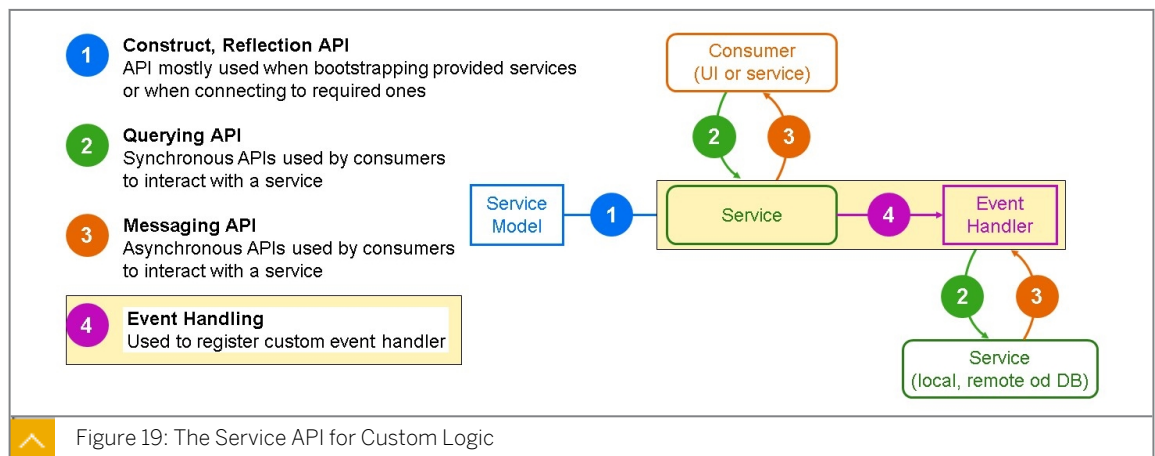


Figure 19: The Service API for Custom Logic

In the figure, The Service API for Custom Logic, you see four types of APIs:

- Construct, Reflection API

This deals with constructing and looking up things in services or connecting to other required services. These are not commonly used. Usually, you will not deal too much with these.

- Querying API

This is a query API, through which you can send synchronous queries to services, including databases.

- Messaging API

This is the asynchronous counterpart of the query API with which services can send messages to one another.

- Event Handling

These are used to register custom event handlers.

As we see on the picture, the box for services appears twice.

This shows that services can mean our own services that we provide as part of the application, or it denotes remote services from other applications or it can mean in databases in which we have the same APIs for in the programming model.

For this unit, we are going to focus on the APIs around registering event handlers to the runtime system (deep dive in the next lesson: Explaining Event Handling in CAP).

Summary

Adding custom business logic to a CAP application allows you to tailor the application to your specific needs and extend its functionality beyond the default capabilities provided by the framework.

Further Reading

Read more about [CAP Core Services](#) here.



LESSON SUMMARY

You should now be able to:

- Explain the need for custom business logic

Describing Error Handling



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain error handling

Error Handling - CAP Service SDK for Node.js

Why We Need This?

Having good error handling is key to ensuring the robustness, correctness, and performance of the given application. Building robust applications requires you to throw and handle exceptions, which occur during the runtime of the application. Therefore, you are introduced to the basic concepts of exception handling in Node.js and the specific techniques for the CAP Service SDK for Node.js.

Error Handling and Error Types

Proper error handling is crucial for today's business applications. Before going into more detail, it is necessary to distinguish between two types of errors:

Programmer Errors

These occur as a result of programming errors (for example, `cannot read 'foo' of undefined`). They must be fixed.

Operational Errors

These occur during runtime (for example, when sending a request to a faulty remote system). They must be handled.

Guidelines

"Let it crash" is a philosophy taken from the Erlang programming language (Joe Armstrong), which is also (partially) applicable to Node.js.

The key takeaways for programming errors are:

- **Fail loudly:** Do not hide errors and continue silently. Ensure to log unexpected errors correctly. Don't catch errors you can't handle.
- **Don't develop in a defensive fashion:** Focus on your business logic and only handle errors when you know they will occur. Use `try/catch` blocks only when necessary.

Never try to catch and handle unexpected errors, rejections of promises, and so on. If it is unexpected, you cannot handle it correctly. If you could, it would be expected (and should already be handled). Even if your apps are stateless, you can never be 100% sure that a shared resource was not affected by the unexpected error. Therefore, you never allow an app to continue running after such an event, especially for multitenant apps where there is a risk of information disclosure.

Following these guidelines will make your code shorter, clearer, and simpler.

Never Hide the Causes of Errors

When an error occurs, it is essential to identify the root cause. The CAP SDK for Node.js, for instance, throws exceptions when certain conditions, such as a CRUD operation violating foreign key constraints, are met. In such cases, the framework raises a specific exception, like `UNIQUE_CONSTRAINT_VIOLATION`. However, if these errors are caught and rethrown without retaining the original details, it becomes challenging to trace and resolve the underlying issue. As a result, the end user is left with only a cryptic error message, making troubleshooting difficult.

For example:

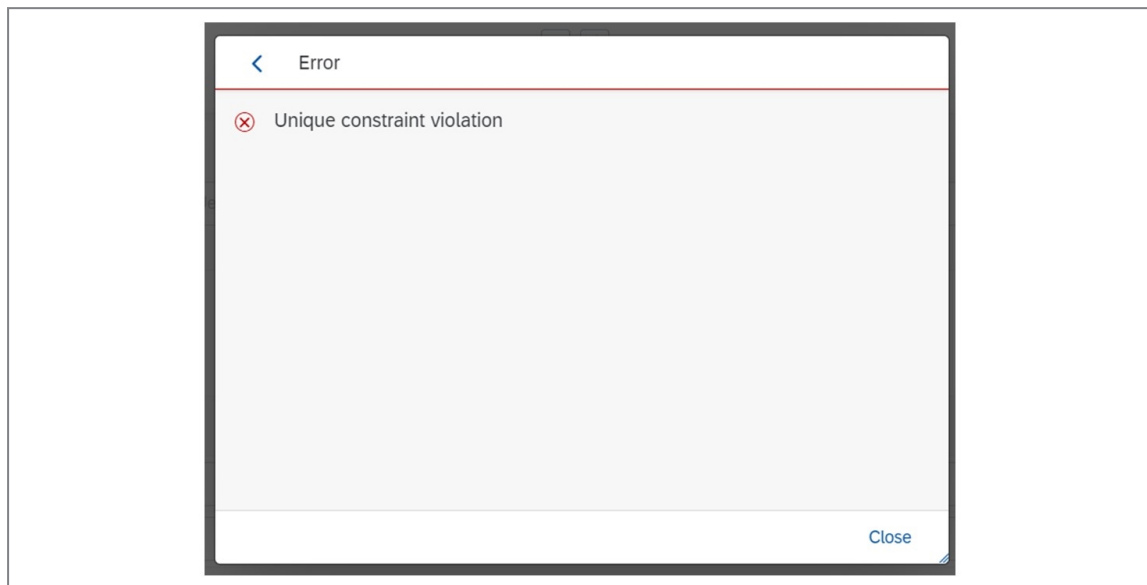


Figure 20: An Exception

```
try {
  // something
} catch (e) {
  // augment instead of replace details
  e.message = 'Oh no! ' + e.message
  e.additionalInfo = 'This is just an example.'
  // re-throw same object
  throw e
}
```

It is therefore useful to provide a meaningful error message.

For this purpose, you can register an error handler in your service implementation, as exemplified in the following snippet.

```
// Imports
const cds = require("@sap/cds");

/**
 * The service implementation with all service handlers
 */
module.exports = cds.service.impl(async function() {
  /**
   * Custom error handler
   *
   * throw a new error with: throw new Error('something bad happened');
   */
});
```

```

*
**/
this.on("error", (err, req) => {
  switch (err.message) {
    case "UNIQUE_CONSTRAINT_VIOLATION":
      err.message = "The entry already exists.";
      break;

    default:
      err.message =
        "An error occurred. Please retry. Technical error message: " +
        err.message;
      break;
  }
});
});

```

This handler now steps in whenever this exception gets triggered and overrides it with an alternative error message:

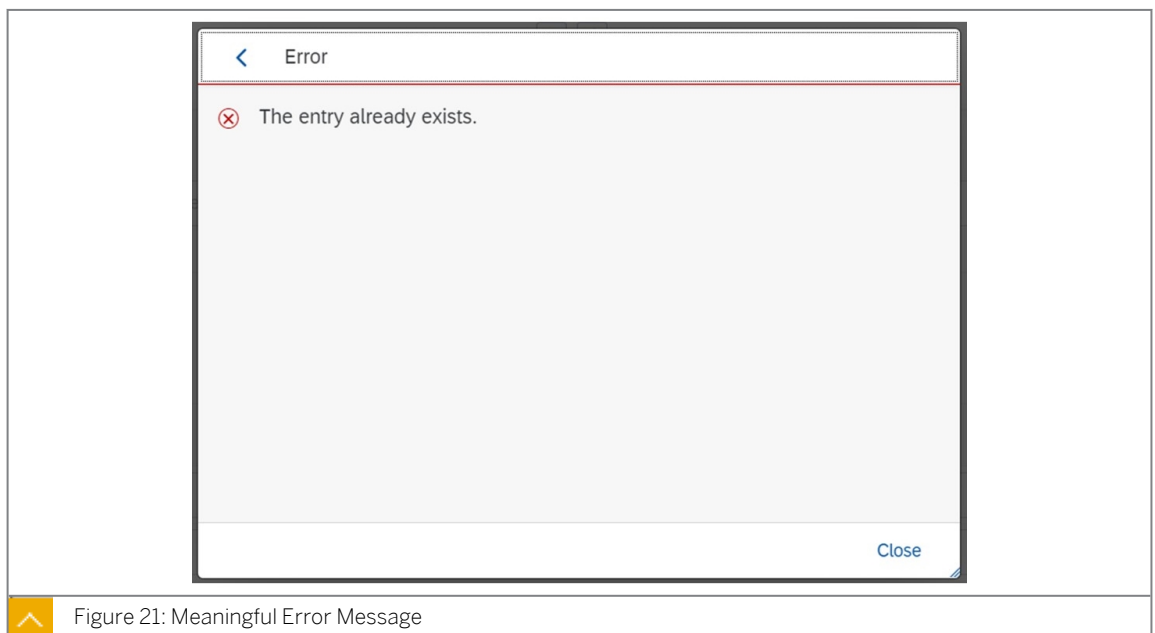


Figure 21: Meaningful Error Message

Raising and Catching Exceptions

You will certainly add your implementations to your services. It is likely that you want to interrupt some operations before something crashes. In this case, you can throw a Node.js exception.

Request Response

You can also use the `req.error()` method to collect messages or errors and return them to the caller in the request-response. Read more [here](#).

```

this.on("submitOrder", async (req) => {
  const { book, amount } = req.data;
  let { stock } = await db.read(Books, book, (b) => b.stock);
  if (stock >= amount) {
    await db.update(Books, book).with({ stock: (stock -= amount) });
    await this.emit("OrderedBook", { book, amount, buyer: req.user.id });
    return req.reply({ stock }); // <-- Normal reply
  } else {
    // Reply with error code 409 and a custom error message
    return req.error(409, `${amount} exceeds stock for book #${book}`);
  }
});

```

```
}  
});
```

Summary

The core error handling concepts in the CAP SDK for Node.js are now familiar to you. We strongly recommend incorporating these concepts to ensure the overall robustness of your CAP application.



LESSON SUMMARY

You should now be able to:

- Explain error handling

Adding Custom Business Logics



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add custom business logic



LESSON SUMMARY

You should now be able to:

- Add custom business logic

Adding Local Launch Pages



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add a local launch page



LESSON SUMMARY

You should now be able to:

- Add a local launch page

Learning Assessment

1. Which pattern do you use to register an event handler?

Choose the correct answer.

☐ A event.()

☐ B phase.()

☐ C srv.()

2. In CAP, which keyword is used to send events?

Choose the correct answer.

☐ A throw

☐ B emit

☐ C actions

☐ D stream

3. How many event handlers can you register for one event phase?

Choose the correct answer.

☐ A Multiple

☐ B Exactly one

☐ C Exactly three

4. Why does CAP set the filerisks-service.js automatically as a handler file for the respective service?

Choose the correct answer.

☐ A Because the filename and the service definition filename are the same.

☐ B Because SAP Business Application Studio added a respective annotation.

5. Which of the following applies to Custom Event handlers?

Choose the correct answers.

- ☐ A Registered custom handlers can add Domain logic to the application.
- ☐ B Multiple Event Handlers can be registered for the same event.
- ☐ C CRUD requests are not served by Generic handlers.
- ☐ D Single handlers can be registered for multiple events.

6. Which are the Service APIs for Custom Logic?

Choose the correct answers.

- ☐ A Messaging API
- ☐ B Event Handling
- ☐ C Querying API
- ☐ D Custom API

7. What can you do to provide meaningful error messages to users in your CAP application?

Choose the correct answer.

- ☐ A Catch exceptions
- ☐ B Register an error handler
- ☐ C Hide the cause of errors
- ☐ D Register an event handler

8. In Node.js, which statement do you use to create an exception?

Choose the correct answer.

- ☐ A throw
- ☐ B try
- ☐ C catch

9. Why does CAP set the file `risks-service.js` automatically as a handler file for the respective service?

Choose the correct answer.

- ☐ A Because the filename and the service definition filename are the same
- ☐ B Because SAP Business Application Studio added a respective annotation

10. Which criticality value is assigned to Negative criticality?

Choose the correct answer.

☐ A 1

☐ B 2

☐ C 3

11. Which is the required status of the IncidentManagement dev space in the exercise Use a Local Launch Page?

Choose the correct answer.

☐ A Running

☐ B Deployed

☐ C New

Learning Assessment - Answers

1. Which pattern do you use to register an event handler?

Choose the correct answer.

☐ A event.()

☐ B phase.()

☒ C srv.()

Correct. You use the srv.() pattern to register an event handler.

2. In CAP, which keyword is used to send events?

Choose the correct answer.

☐ A throw

☒ B emit

☐ C actions

☐ D stream

Correct. In CAP, you use the emit keyword to send events.

3. How many event handlers can you register for one event phase?

Choose the correct answer.

☒ A Multiple

☐ B Exactly one

☐ C Exactly three

Correct. You can register multiple event handlers for one event phase.

4. Why does CAP set the `filerisks-service.js` automatically as a handler file for the respective service?

Choose the correct answer.

- ☒ A Because the filename and the service definition filename are the same.
- ☐ B Because SAP Business Application Studio added a respective annotation.

Correct. CAP sets the `filerisks-service.js` automatically as a handler file for the respective service because the filename and the service definition filename are the same.

5. Which of the following applies to Custom Event handlers?

Choose the correct answers.

- ☒ A Registered custom handlers can add Domain logic to the application.
- ☒ B Multiple Event Handlers can be registered for the same event.
- ☐ C CRUD requests are not served by Generic handlers.
- ☒ D Single handlers can be registered for multiple events.

Correct. The following applies to Custom Event handlers: Registered custom handlers can add Domain logic to the application, multiple Event Handlers can be registered for the same event, and single handlers can be registered for multiple events.

6. Which are the Service APIs for Custom Logic?

Choose the correct answers.

- ☒ A Messaging API
- ☒ B Event Handling
- ☒ C Querying API
- ☐ D Custom API

Correct. The Service APIs for Custom Logic are: Messaging API, Event Handling, and Querying API.

7. What can you do to provide meaningful error messages to users in your CAP application?

Choose the correct answer.

- ☐ A Catch exceptions
- ☒ B Register an error handler
- ☐ C Hide the cause of errors
- ☐ D Register an event handler

Correct. To provide meaningful error messages to users in your CAP application, you can register an error handler.

8. In Node.js, which statement do you use to create an exception?

Choose the correct answer.

- ☒ A throw
- ☐ B try
- ☐ C catch

Correct. In Node.js, you use the throw statement to create an exception.

9. Why does CAP set the file `risks-service.js` automatically as a handler file for the respective service?

Choose the correct answer.

- ☒ A Because the filename and the service definition filename are the same
- ☐ B Because SAP Business Application Studio added a respective annotation

Correct. CAP sets the file because the filename and the service definition filename are the same.

10. Which criticality value is assigned to `Negative` criticality?

Choose the correct answer.

- ☒ A 1
- ☐ B 2
- ☐ C 3

Correct. The criticality value 1 is assigned to `Negative` criticality.

11. Which is the required status of the IncidentManagement dev space in the exercise Use a Local Launch Page?

Choose the correct answer.

- ☒ **A** Running
- ☐ **B** Deployed
- ☐ **C** New

Correct. The required status is Running.

UNIT 5

Understanding Authorization and Trust Management

Lesson 1

Describing Authorization and Trust Management (XSUAA)

77

Lesson 2

Defining CDS Restrictions and Roles

83

UNIT OBJECTIVES

- Describe the SAP Authorization and Trust Management service
- Define CDS restrictions and roles

Describing Authorization and Trust Management (XSUAA)



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the SAP Authorization and Trust Management service

SAP Authorization and Trust Management Service (XSUAA)

Business Case

Your company is developing a set of applications on the SAP Business Technology Platform, Cloud Foundry environment. As the applications are running in the cloud, you want to protect them from unauthorized access. The SAP Authorization and Trust Management service in conjunction with the Extended Services - User Account and Authentication (XSUAA) service provide all the required features.

Manage User Authorizations

The [SAP Authorization and Trust Management service](#) lets you manage user authorizations and trust to identity providers. Identity providers are the user base for applications. You can use an identity authentication tenant, an SAP on-premise system, or a custom corporate identity provider. User authorizations are managed using technical roles at the application level, which can be aggregated into business-level groups and role collections for large-scale cloud scenarios.

Platform Users Versus Business Users

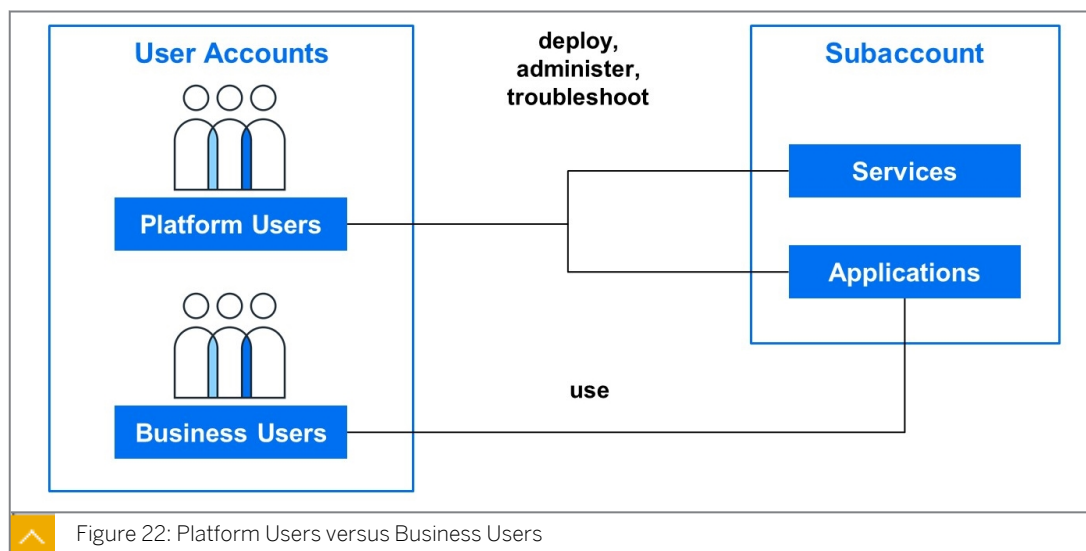
Platform Users

Platform users are usually developers, administrators, or operators who deploy, administer, and troubleshoot applications and services on SAP BTP.

For platform users, the default identity provider is SAP ID service.

Business Users

Business users use the applications that are deployed to SAP BTP. For example, the end users of your deployed application or users of subscribed apps or services, such as SAP Business Application Studio or SAP Web IDE, are business users.

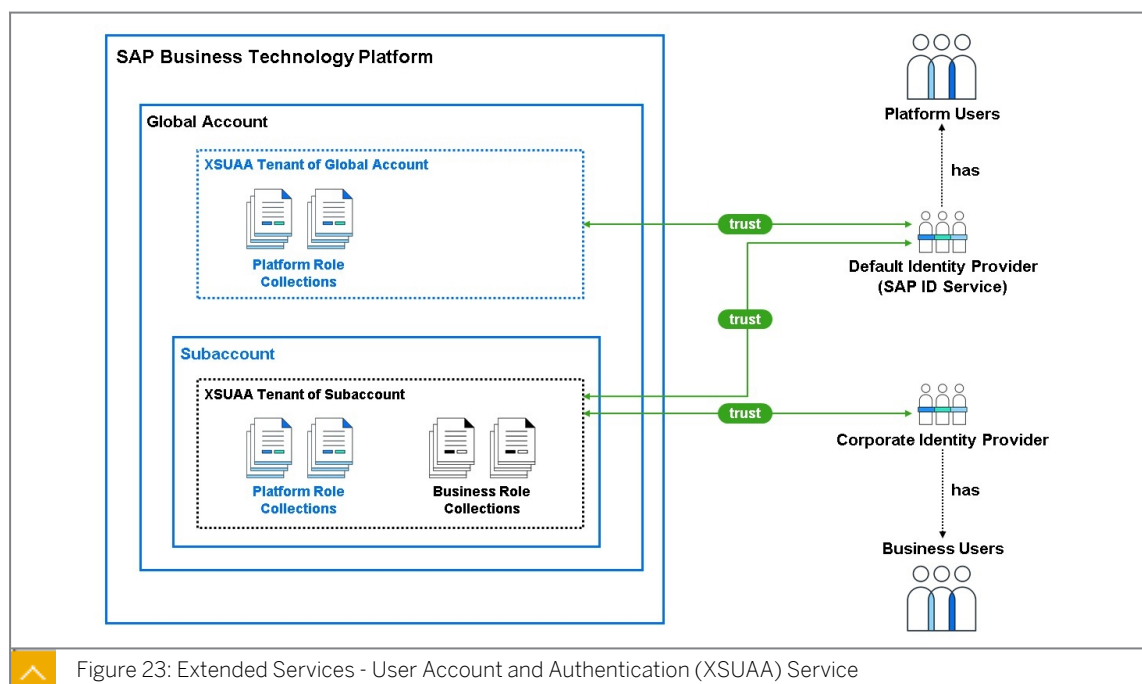


Extended Services - User Account and Authentication (XSUAA) Service

The Extended Services - User Account and Authentication (XSUAA) service is one of the most important components to deal with when developing your own applications on Cloud Foundry. It authenticates and authorizes your users and assigns the right privileges to your user's session so your application can do the following:

- Identify the user by Email, UserId, First, and Lastname.
- Check its roles (scopes) to decide if a user is allowed to do something or prohibit its action.

The XSUAA is an internal development of SAP. SAP took the base of the [open source UAA OAuth2 Provider of Cloud Foundry](#) and extended it with SAP specific features to be used in SAP applications. One important thing is that the XSUAA does NOT store "real" users. This is why the XSUAA needs to trust an external Identity Provider (IdP).



**Note:**

This graphic only applies to SAP Business Technology Platform cloud management tools Feature Set B. Check out [User and Member Management](#) in the SAP Help Portal pages to better understand the differences between Feature Set A and Feature Set B in regard to user management.

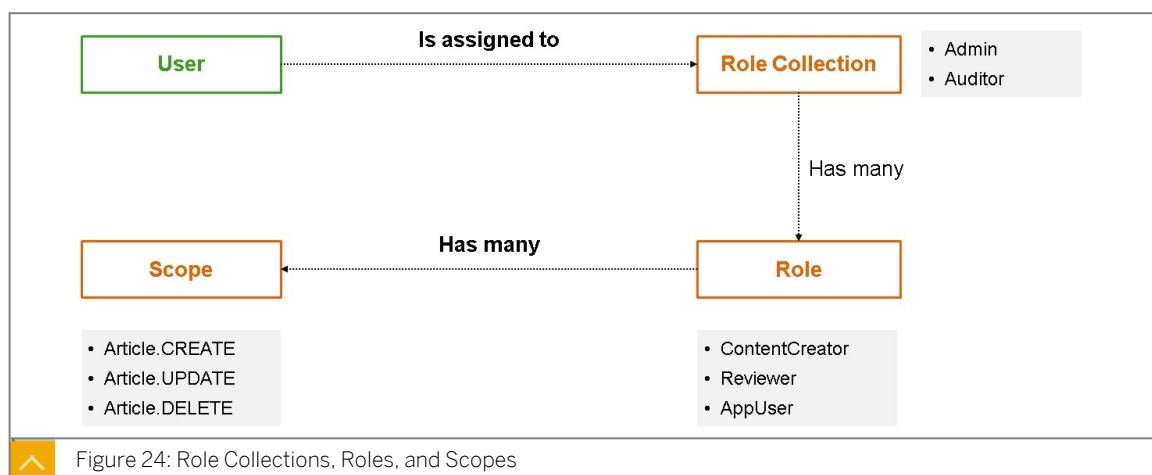
In Feature Set B, your SAP BTP global account has its own XSUAA tenant. This XSUAA tenant by default has a trust relationship to the SAP ID service. The SAP ID service manages a large base of users, that have created a [user account with SAP](#). You can add users that exist in the SAP ID service as members to your global account and subaccount. For these users to be able to perform administrative tasks, they need to be assigned with corresponding role collections. There is a set of default platform role collections, like Global Account Administrator, Global Account Viewer, Subaccount Administrator, or Cloud Connector Administrator that you can use for the purpose of assigning SAP BTP account management authorizations. See this [SAP Help Portal page](#) for further information on the default role collections for account management.

Contrary to the platform users from the SAP ID service, your business users can also be provided via your own corporate identity provider. These are the users you want to provide access to your business applications. These might be SaaS applications provided by SAP, like SAP Business Application Studio or SAP Workflow Management, or your own applications that you develop on the SAP BTP. The business applications have their own role collections, like - for example, Business_Application_Studio_Developer, WorkflowManagementAdmin, or any custom role collection that you create for your application.

The following sections shed more light on the terms role collection, role, and scope, and their relationship to each other.

Role Collections, Roles, and Scopes

The following figure, Role Collections, Roles, and Scopes, shows the relationships between role collections, roles, and scopes.



Scopes

Scopes are arbitrary values that express authorizations / access rights in an application or service. Scopes need to be prefixed with an `xsappname` to make them uniquely identifiable.

Roles

Roles are entities that hold several scopes. Scopes can be put in multiple roles, so you are not limited to have scopes sitting in just one role.

Role Collections

Role collections contain one or more roles. A role can be used in multiples. But it is totally fine to have, for example, a called `Admin` that only has an `admin` role.

Role collections are stored as an assignment in the XSUAA and are THE entity that can be assigned to a certain business user.

How Does it Work in Practice?

The following section is a short summary of the [official help page](#).

In the diagram, you can see that there are different personas. One persona is the developer working within a project and space. The other persona is an admin taking care of the CF account as a security admin.

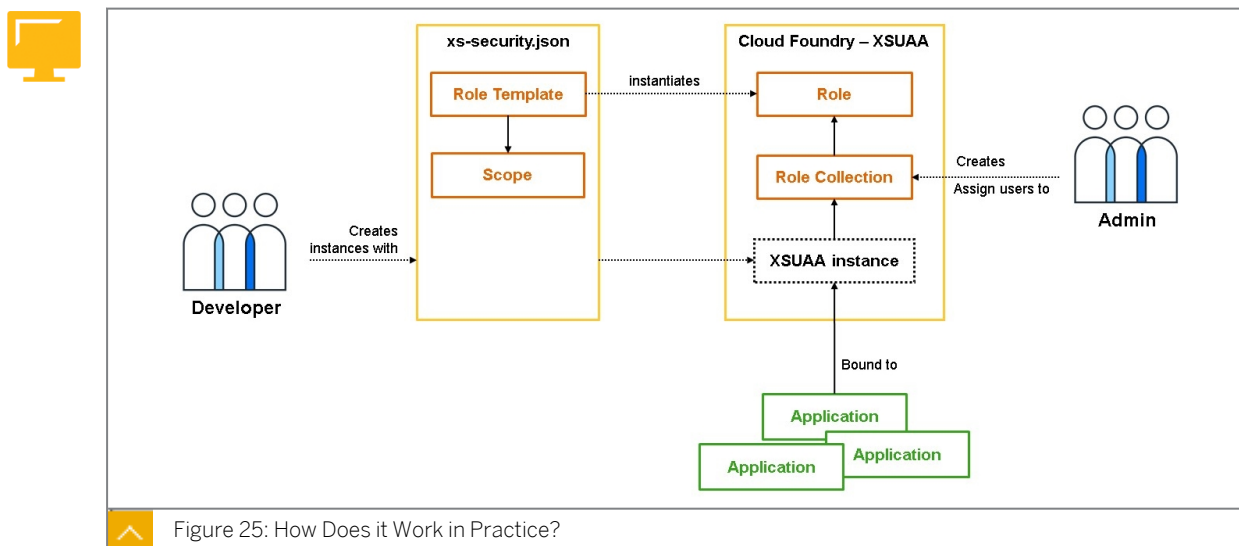


Figure 25: How Does it Work in Practice?

When you, as a developer, build a new business application, you define scopes and pre-bundle them in role-templates. You perform these definitions in the so called application security descriptor (`xs-security.json`) file. You use the `xs-security.json` file to create an instance of the XSUAA service, which is bound to the corresponding business application(s). The role-template definitions translate into roles. You, as an administrator, assemble these roles into s and assign them to the business users of your application.

What is `xs-security.json`?

To simplify things, let's just call the `xs-security.json` the "declaration of your app's security". The following `xs-security.json` is an example for the `incident-management` application:

```
{
  "xsappname": "incident-management",
  "tenant-mode": "dedicated",
  "scopes": [
```

```

    {
      "name": "$XSAPPNAME.IncidentViewer",
      "description": "IncidentViewer"
    },
    {
      "name": "$XSAPPNAME.IncidentManager",
      "description": "IncidentManager"
    }
  ],
  "attributes": [],
  "role-templates": [
    {
      "name": "IncidentViewer",
      "description": "generated",
      "scope-references": [
        "$XSAPPNAME.IncidentViewer"
      ],
      "attribute-references": []
    },
    {
      "name": "IncidentManager",
      "description": "generated",
      "scope-references": [
        "$XSAPPNAME.IncidentManager"
      ],
      "attribute-references": []
    }
  ]
}

```

You have to tell the XSUAA service how to call your application (`xsappname`) and further define your scopes and role-templates. This can also be defined in the `mta.yaml` file as well. The scopes are being used within the application to check concrete permissions whenever a user tries to perform a certain action.

Summary

You have now gained a general understanding of the SAP Authorization and Trust Management service, the relevance of the Extended Services - User Account and Authentication (XSUAA) service, and the basic concepts of user management and assignment. You know what the `xs-security.json` application security descriptor file is used for and how it relates to scopes, roles, and the XSUAA service. You also know where to look for additional information.

Further Reading

- Parts of the text and graphics were taken from the blog post [Demystifying XSUAA in SAP Cloud Foundry](#). Take a look at the blog for even more information. The explanations in regard to user management in the blog post refer to the SAP BTP cloud management tools, Feature Set A.
- [The Application Security Descriptor](#)



LESSON SUMMARY

You should now be able to:

- Describe the SAP Authorization and Trust Management service

Defining CDS Restrictions and Roles



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define CDS restrictions and roles



LESSON SUMMARY

You should now be able to:

- Define CDS restrictions and roles

Learning Assessment

1. Which user types work on and with SAP BTP?

Choose the correct answers.

- ☐ A Professional users
- ☐ B Platform users
- ☐ C Trial users
- ☐ D Business users

2. What is the default identity provider for SAP BTP platform users?

Choose the correct answer.

- ☐ A SAP Cloud Identity Services
- ☐ B SAP ID service
- ☐ C SAML 2.0
- ☐ D XSUAA

3. What does the Extended Services - User Account and Authentication (XSUAA) service enable your app to do?

Choose the correct answers.

- ☐ A Store "real" users
- ☐ B Identify users by address and social security ID
- ☐ C Identify users by e-mail, userId, first and last name
- ☐ D Check users' roles to allow or prohibit actions

4. Which file contains an app's "declaration of security"?

Choose the correct answer.

- ☐ A xs-sec.json
- ☐ B xs-app.json
- ☐ C xs-security.json

5. How do you add authorization and trust management to your CAP project?

Choose the correct answer.

- ☐ A cds add security
- ☐ B cds add uaa
- ☐ C cds add xsuaa

6. Which files do you modify to store project configurations?

Choose the correct answers.

- ☐ A .cdsrc.json
- ☐ B passport.js
- ☐ C package.json

Learning Assessment - Answers

1. Which user types work on and with SAP BTP?

Choose the correct answers.

- ☐ A Professional users
- ☒ B Platform users
- ☐ C Trial users
- ☒ D Business users

Correct. Platform users and business users work on and with SAP BTP.

2. What is the default identity provider for SAP BTP platform users?

Choose the correct answer.

- ☐ A SAP Cloud Identity Services
- ☒ B SAP ID service
- ☐ C SAML 2.0
- ☐ D XSUAA

Correct. SAP ID service is the default identity provider for SAP BTP platform users.

3. What does the Extended Services - User Account and Authentication (XSUAA) service enable your app to do?

Choose the correct answers.

- ☐ A Store "real" users
- ☐ B Identify users by address and social security ID
- ☒ C Identify users by e-mail, userId, first and last name
- ☒ D Check users' roles to allow or prohibit actions

Correct. XSUAA enables your app to identify users by e-mail, userId, first and last name, and check users' roles to allow or prohibit actions.

4. Which file contains an app's "declaration of security"?

Choose the correct answer.

- ☐ A xs-sec.json
- ☐ B xs-app.json
- ☒ C xs-security.json

Correct. The file xs-security.json contains an app's "declaration of security".

5. How do you add authorization and trust management to your CAP project?

Choose the correct answer.

- ☐ A cds add security
- ☐ B cds add uaa
- ☒ C cds add xsuaa

Correct. You can install Passport.js to add authentication to your CAP application.

6. Which files do you modify to store project configurations?

Choose the correct answers.

- ☒ A .cdsrc.json
- ☐ B passport.js
- ☒ C package.json

Correct. You modify .cdsrc.json and package.json files to store project configurations.

UNIT 6

Consuming External Services

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UNIT OBJECTIVES

- Explain the extensibility and connectivity capabilities in CAP
- Explaining Unit Tests
- Add test cases
- Add and consume an external service

Explaining Extensibility and Connectivity in CAP



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the extensibility and connectivity capabilities in CAP

Extensibility and Connectivity Capabilities in CAP

Usage Scenario

In CAP, extensibility and connectivity are fundamental concepts that enable developers to build flexible and connected applications that can adapt to changing business needs and integrate seamlessly with various data sources and services.

Extensibility in CAP

Extensibility in CAP refers to the ability to customize and extend the data models, business logic, and user interfaces of applications to accommodate specific business requirements. CAP offers several mechanisms for achieving extensibility:

Data Model Extensibility

Developers can define data models using Core Data Services (CDS). CAP allows you to add custom entities, fields, and relationships to the data model to meet specific business needs. You can extend existing data models without altering the core structure, making it easier to adapt to changing data requirements.

Behavioral Extensibility

CAP provides service handlers and event handlers that allow you to define custom business logic. You can implement custom code that executes during various application events, such as create, read, update, and delete (CRUD) operations, validations, and data transformations.

UI Extensibility

When building user interfaces (UIs) with CAP, you can customize and extend SAP Fiori applications. This includes adding new UI components, modifying existing views, or creating custom pages to tailor the user experience to your organization's needs.

Service Extensibility

CAP allows you to create custom OData services that expose additional functionality or data to external consumers. These services can integrate with other SAP applications or external systems, facilitating data exchange and interoperability.

Localization

CAP supports localization features that enables you to adapt your application to different languages and regions, which is crucial for global deployments.

Connectivity in CAP

Connectivity in CAP is the capability to seamlessly connect and integrate with various data sources, services, and SAP internal or external systems. CAP provides multiple mechanisms for achieving connectivity:

Database Connectivity

CAP abstracts database interactions, allowing you to work with data models using Core Data Services (CDS) without being concerned about the underlying database system. CAP supports various database systems, including SAP HANA, SQLite, H2 (Java only), PostgreSQL, providing flexibility in choosing the right database for your application.

Service Connectivity

CAP applications can interact with external services and APIs using HTTP-based calls, REST, GraphQL, or other protocols. This enables integration with third-party systems and the consumption of external data. In this unit, we are going to deep dive on the integration with an external service.

Event-based Connectivity

CAP supports event-driven architecture, allowing your application to emit and listen for events. This facilitates real-time data synchronization and communication between different parts of your application or with external systems.

Message Queues and Event Brokers

CAP can integrate with message queuing systems and event brokers like SAP Enterprise Messaging or third-party solutions. This enables asynchronous communication and event-driven processing, which is essential for scalable and responsive applications.

SAP Integration

CAP can easily integrate with other SAP solutions, such as SAP Integration Suite (Cloud Integration, API Management). This allows you to create end-to-end business processes that span multiple SAP applications and services.

Security and OAuth Integration

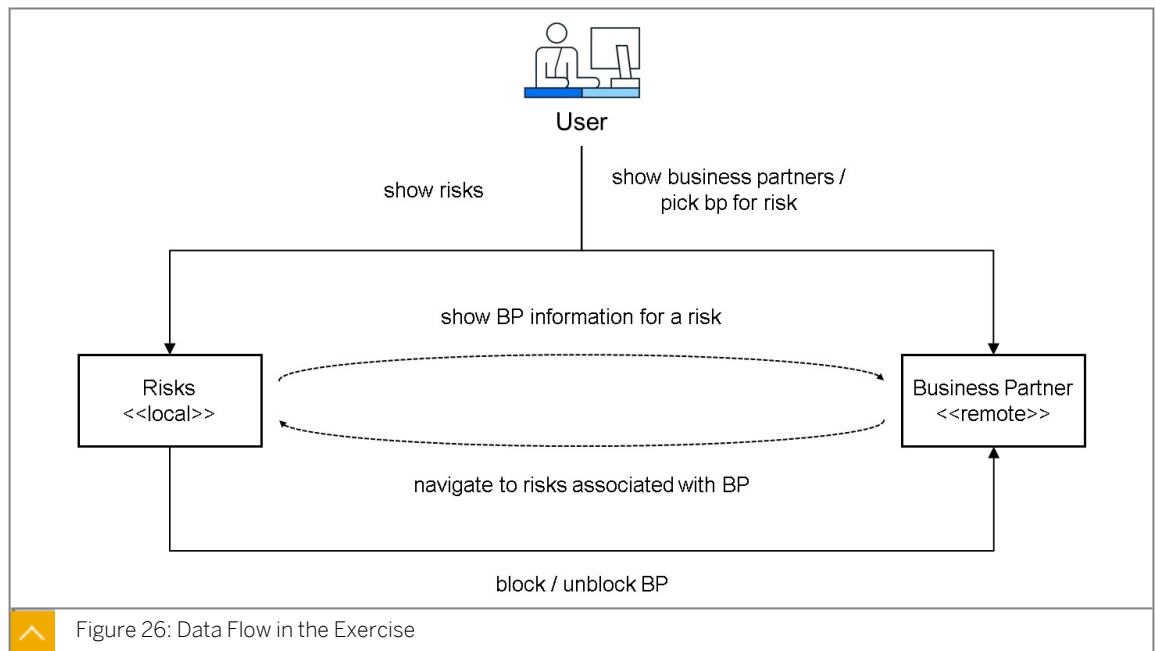
CAP provides built-in support for authentication and authorization, including OAuth 2.0. This makes it easier to secure your application and integrate it with identity providers or SAP's authentication services.

Addition of the Consumption of an External Service to Your CAP Application

SAP Business Accelerator Hub

SAP Business Accelerator Hub is a central repository and platform provided by SAP that offers a comprehensive catalog of SAP and non-SAP APIs, including documentation, specifications, and resources to help developers discover, explore, and consume APIs effectively.

In this unit's exercise, for our risk management application we want to add the business partner information by calling the external data source (SAP S/4HANA Cloud system) to read the risks and mitigations information alongside the business partners. The application helps to reduce risks associated with business partners using a remote API Call to SAP S/4HANA Cloud. To get the data for the business partner, you must connect the service to the Sandbox environment of the SAP Business Accelerator Hub (deep dive in the exercise).



To communicate to remote services, CAP must know their definitions. These definitions are made available by the service provider. As they aren't defined within your application but imported from outside, they're called external service APIs in CAP. Currently, EDMX files for OData V2 and V4 are supported.

The steps for using the Business Partner external API in the risk management project are as follows:

1. Connect your app with the Business Partner API Sandbox Environment of the SAP API Business Hub. For this, you can configure an application defined destination in Node.js by just adding this configuration on the package.json file.

```

"cds": {
  "requires": {
    "API_BUSINESS_PARTNER": {
      "kind": "odata",
      "model": "srv/external/API_BUSINESS_PARTNER",
      "credentials": {
        "url": "https://sandbox.api.sap.com/s4hanacloud/sap/opu/odata/sap/API_BUSINESS_PARTNER/"
      }
    }
  }
}

```

2. Consume the External Service in your UI application.
3. Add the Business Partner Field to the UI.

To manage the connectivity to external systems or services, you can choose to use destinations: SAP BTP destination of application defined destinations (previous example). They provide a way to define and configure the connection details to these external systems, making it easier to handle various aspects of integration, such as authentication, endpoint URLs and more, by abstracting the underlying connection information.

In using SAP BTP destination, the application must be deployed to BTP. This is because we require instances of SAP's Destination Service and Connectivity Service (in a premise external service), which can only be created and bound to an application on BTP. The

Connectivity service provides a connectivity proxy that you can use to access on-premise resources. Using the Destination service, you can retrieve and store the technical information about the target resource (destination) that you must connect your application to a remote service or system. You can use both services together and separately, depending on the needs of your specific scenario.

If the external system is on-premise, the Cloud Connector must be configured in order to use the on-premise resources.

Summary

You now have a high-level understanding of the extensibility and connectivity possibilities using the CAP framework.

Further Reading

For more information, consult the following:

- [The CAP Cookbook](#).
- [Consuming Services](#).
- [Using Destinations](#).
- [Cloud Connector](#).
- [BTP Connectivity](#).



LESSON SUMMARY

You should now be able to:

- Explain the extensibility and connectivity capabilities in CAP

Explaining Unit Tests



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explaining Unit Tests

Explaining Unit Tests

Usage Scenario

Imagine you are developing a Node.js backend for a large e-commerce platform. The platform handles various tasks like user registration, product listings, order processing, and payments. Each of these functionalities involves a lot of logic, and any bug in one part can affect the entire system. Manually testing each piece of functionality every time a new feature is added is both tedious and error-prone. This is where **unit testing** comes into play.

What Is Unit Testing?

Unit testing involves testing individual functions or methods of your codebase in isolation. The goal is to ensure that each unit behaves as expected under different conditions. Unit tests can be characterized through the following:

- **Isolated:** A unit test only tests one piece of code in isolation from external systems (for example, databases, APIs).
- **Fast:** Unit tests run quickly to allow for rapid feedback during development.
- **Repeatable:** The tests always return the same result for the same input.

Why Is Unit Testing Important?

Early detection of bugs: Unit tests help developers catch bugs early in the development cycle, reducing the cost and effort required to fix them.

- **Confidence in code changes:** By running unit tests frequently, you can confidently refactor your code or add new features without worrying about breaking existing functionality.
- **Automated testing:** Unit tests can be automated and integrated into your CI/CD pipelines, reducing the need for manual testing.
- **Improved code quality:** Writing unit tests often leads to cleaner, more modular, and maintainable code.

Example: Writing Unit Tests in Node.js

In Node.js, the most common libraries used for unit testing are **Mocha** (test framework) and **Chai** (assertion library). Let's look at an example:

We will write a test for a basic add function using Mocha and Chai.

1. Install Mocha and Chai:

```
npm install mocha chai --save-dev
```

2. Create a file `math.js` with the following content:

```
function add(a, b) {  
  return a + b;  
}  
  
module.exports = { add };
```

3. Write the unit test in a separate file `test/math.test.js`.

```
const { expect } = require('chai');  
const { add } = require('../math');  
  
describe('Math Functions', () => {  
  it('should return the sum of two numbers', () => {  
    const result = add(2, 3);  
    expect(result).to.equal(5);  
  });  
});
```

4. To run the tests, add the following script to `package.json`

```
"scripts": {  
  "test": "mocha"  
}
```

5. Run the test using the following command.

```
npm test
```

This example shows a simple unit test using Mocha as the test runner and Chai for assertions.

Summary

Unit testing is a crucial practice in Node.js development for verifying the correctness of individual functions and methods. By writing isolated, repeatable, and automated tests, you can ensure the reliability of your application while improving code quality. Tools like Mocha or Chai make unit testing in Node.js straightforward and efficient.

Further Reading

- [Mocha Documentation](#)
- [Chai Assertion Library](#)



LESSON SUMMARY

You should now be able to:

- Explaining Unit Tests

Adding Test Cases



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add test cases



LESSON SUMMARY

You should now be able to:

- Add test cases

Adding External Services



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add and consume an external service



LESSON SUMMARY

You should now be able to:

- Add and consume an external service

Learning Assessment

1. When you develop a CAP application, what can be used to connect to an external Cloud system?

Choose the correct answers.

- ☐ A SAP BTP destinations using Destination Service
- ☐ B Cloud Connector
- ☐ C Node.js configured destinations
- ☐ D Global Account Destinations

2. Which of the following is a key characteristic of unit testing?

Choose the correct answer.

- ☐ A Testing the individual components of code in isolation
- ☐ B Testing the performance of the entire application
- ☐ C Testing the integration between different systems

3. Which library is commonly used for assertions in Node.js unit testing?

Choose the correct answer.

- ☐ A Chai
- ☐ B Express
- ☐ C Passport

4. In the exercise Add Test Cases, you use the file: test_main.js.

Determine whether this statement is true or false.

- ☐ True
- ☐ False

5. What do you need to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use?

Choose the correct answer.

- ☐ A An API key
- ☐ B A personal token

6. What does the .env file provide?

Choose the correct answer.

- ☐ A Values into the runtime environment of a CAP service
- ☐ B Values for your version-management-system

Learning Assessment - Answers

1. When you develop a CAP application, what can be used to connect to an external Cloud system?

Choose the correct answers.

- ☒ A SAP BTP destinations using Destination Service
- ☒ B Cloud Connector
- ☒ C Node.js configured destinations
- ☐ D Global Account Destinations

Correct. You can use the following: SAP BTP destinations using Destination Service, the Cloud Connector, and Node.js configured destinations.

2. Which of the following is a key characteristic of unit testing?

Choose the correct answer.

- ☒ A Testing the individual components of code in isolation
- ☐ B Testing the performance of the entire application
- ☐ C Testing the integration between different systems

Correct. A key characteristic of unit testing is testing the individual components of code in isolation.

3. Which library is commonly used for assertions in Node.js unit testing?

Choose the correct answer.

- ☒ A Chai
- ☐ B Express
- ☐ C Passport

Correct. The Chai library is used for assertions in Node.js unit testing.

4. In the exercise Add Test Cases, you use the file: test_main.js.

Determine whether this statement is true or false.

☐ True

☐ False

Correct. In the tests folder, you use the file test.js.

5. What do you need to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use?

Choose the correct answer.

☒ A An API key

☐ B A personal token

Correct. You need an API key to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use.

6. What does the .env file provide?

Choose the correct answer.

☒ A Values into the runtime environment of a CAP service

☐ B Values for your version-management-system

Correct. The .env file provides values into the runtime environment of a CAP service.

UNIT 7

Deploying the Application

Lesson 1

Identifying Deployment Options in CAP

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Lesson 2

Explaining the Deployment Process

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Lesson 3

Preparing for Deployment in the SAP BTP, Cloud Foundry Runtime

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Lesson 4

Deploying the Application

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UNIT OBJECTIVES

- Identify the deployment options of a CAP application
- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file
- Prepare the application for deployment
- Deploying the application

Identifying Deployment Options in CAP



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify the deployment options of a CAP application

Deployment Options in CAP

Introduction

You have developed a stable version of your CAP project and now you want to deploy it to the SAP BTP. You have different options to deploy your project. In this lesson, you will learn about the different deployment options.

Deployment Options

When deploying a CAP application, there are different options. The following figure, Deployment Options of a CAP Application, shows the different options.

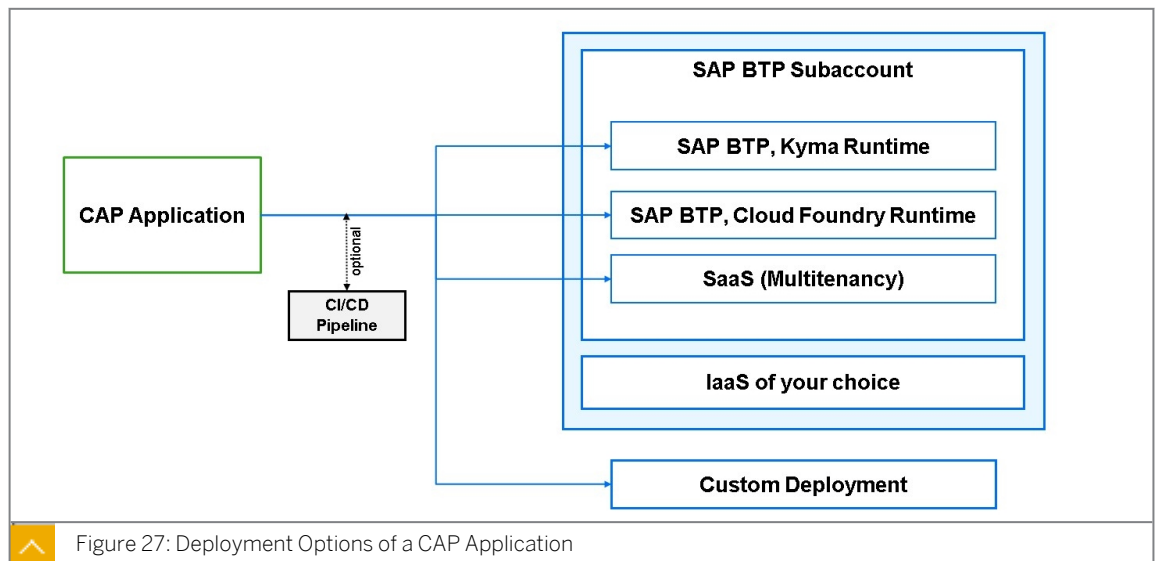


Figure 27: Deployment Options of a CAP Application

On the SAP Business Technology Platform (SAP BTP), the two runtimes SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime can be targeted for deployment. As well as deploying them in runtime on a specific subaccount, it is also possible to deploy the CAP application as a Service (SaaS) to SAP BTP, Cloud Foundry Runtime, or SAP BTP, Kyma environment. The application can then be subscribed by multiple SaaS consumers (other subaccounts) from the Service Marketplace.

The [SAP BTP Developer's Guide](#) also provides an overview of the different approaches available on SAP BTP.

SAP BTP, Cloud Foundry Runtime

The deployment to Cloud Foundry is straightforward. CAP uses the **Multitarget Application (MTA)** approach to deploy the application to the Cloud Foundry environment. A **MTA** is logically a single application consisting of multiple related and interdependent parts (modules) that are developed using different technologies or programming paradigms and designed to run on different target runtime environments with a single, consistent life cycle.

More on MTA later in this course. See the CAP Documentation for a complete [Cloud Foundry Deployment Guide](#).

SAP BTP, Kyma Runtime

Deployment to the SAP BTP, Kyma Runtime is possible. CAP uses [Pack](#) and [Helm](#) Charts to ease the deployment process. You don't have to deal with creating your own Dockerfiles and container images. The tool-set will handle this for you. If you are new to Kyma and Kubernetes, make sure to check out the free [Kyma Learning Journey](#).

Read more about the [Kyma deployment process](#) in the CAP documentation.

Multitenancy (SaaS)

Deploying a CAP (SAP Cloud Application Programming Model) application as a Software as a Service (SaaS) application involves making your application accessible to multiple tenants within a shared infrastructure. These applications are designed to provide data isolation, customization, security, and scalability for each tenant while efficiently using shared resources. This approach is common in software-as-a-service (SaaS) and cloud-based enterprise applications, where multiple customers or organizations use the same application instance while maintaining their data and configurations separately.

CAP has built-in support for multitenancy with the `@sap/cds-mtxs` library, providing a set of CAP services which implement multitenancy, features, toggles and extensibility.

The steps for enabling multitenancy for your SaaS application are:

1. `npm add @sap/cds-mtxs`
2. Navigate to `package.json` and add one or more of the following flags, depending on the requirements:

```
"cds": {
  "requires": {
    "multitenancy": true,
    "extensibility": true,
    "toggles": true
  }
}
```

SaaS applications must register with the SAP BTP SaaS Provisioning service to allow different tenants to subscribe or unsubscribe to the application. On top of this, the Service Manager can be used for creating a new SAP HANA Deployment Infrastructure (HDI) container for each tenant and for retrieving tenant-specific database connections.

As this is a multitenant application, using a Standalone App Router is best practice for creating multitenant applications on SAP BTP. The App Router is a Node.js library used as a single entry point for an application running on SAP BTP. More details on this on the next lesson.

Read more about [Building SAAS Applications with CAP](#) and [MTX Services](#) in the CAP documentation.

Custom Deployment

Creating a custom build of an SAP Cloud Application Programming (CAP) application for deployment typically involves the process of compiling and packaging your CAP application's source code and assets into a production-ready bundle that can be deployed to your chosen deployment environment.

`cds build` executes build tasks on your project folders to prepare them for deployment. You can customize the behavior of the `cds build` process by providing options, flags, or configuration files as part of your project's settings or through command-line arguments.

An example of a command with options: `cds build --production`.

Read more about [Custom Builds for deployment](#) in the CAP documentation.

Summary

When deploying an SAP Cloud Application Programming (CAP) application, you have several deployment options, depending on your specific requirements, infrastructure, and preferences. If SAP BTP is the cloud platform of choice, the environments that can be used for both multitenant (SAAS) and single tenant application on the Cloud Foundry or Kyma environments. Choose Cloud Foundry when you need a simpler deployment model with high-level abstractions. Choose Kyma when you require extensive control over containerized applications and microservices, especially in a Kubernetes ecosystem.



LESSON SUMMARY

You should now be able to:

- Identify the deployment options of a CAP application

Explaining the Deployment Process



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file

The Deployment Process

Usage Scenario

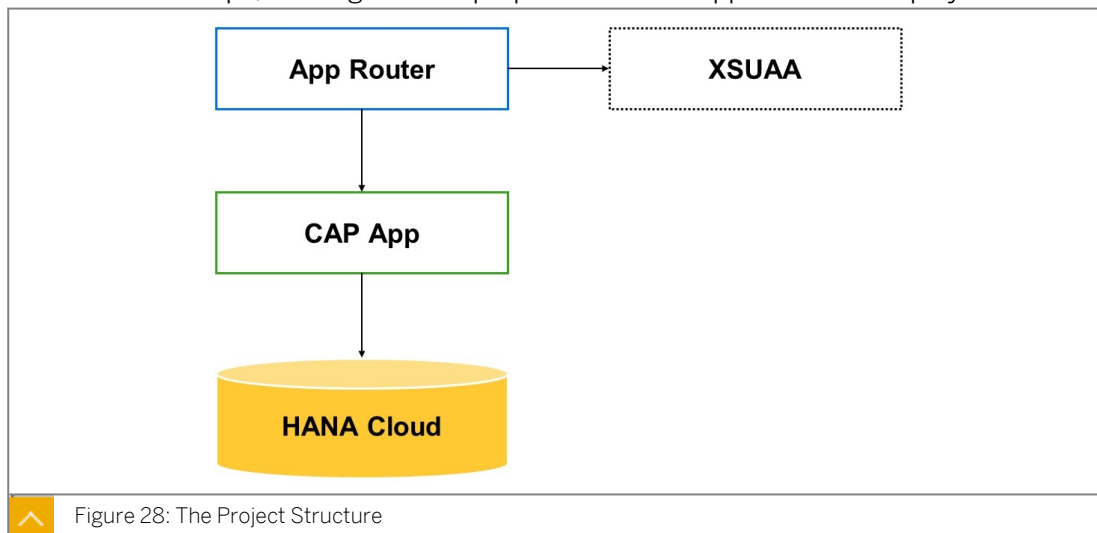
Deploying a CAP application on SAP Business Technology Platform involves several steps, including setting up the required services, configuring the deployment environment, and deploying the application itself.

1. Set up SAP BTP Environment:
 - Log in to the SAP BTP Cockpit.
 - Set up the necessary subaccounts and entitlements for the services you need.
 - Make sure the necessary entitlements for the Cloud Foundry environment are available.
2. Prepare Your CAP Application:
 - Create your CAP project.
 - Ensure the application is ready for deployment and has the necessary configurations for the target environment.
 - Configure Deployment Settings in the mta.yaml file.
 - Define the required services and their bindings in the deployment descriptor.
3. Build the Application:
 - Use the CAP tooling or the appropriate build tools to build the application.
 - Ensure that the application builds successfully without any errors or warnings.
4. Deploy the Application:
 - Use the Cloud Foundry CLI or the SAP Business Application Studio to deploy the application to SAP BTP.
 - Monitor the deployment process for any errors or issues that might arise.

By following these steps, you can effectively deploy a CAP application on SAP BTP and ensure its smooth operation within the SAP cloud environment.

Deployment Process

1. Considering step 1 in place (Access to SAP BTP, Cloud Foundry environment, a CF space created and a SAP HANA Cloud database instance configured and running), the focus will be on the next steps, starting with the preparation of the application for deployment.



2. Prepare Your CAP Application:

Prerequisites for preparing the application for deployment:

- a. SAP HANA Cloud database is configured in our CAP project: `cds add hana --for production`
- b. Authentication and authorization is configured by using the SAP Authorization and Trust Management Service: `cds add xsuaa --for production`. This generates an `xs-security.json` file, with roles/scopes derived from authorization-related annotations in the CDS models created. Ensure to rerun `cds compile --to xsuaa` in case of changes to the annotations in the project.
- c. Using Multitarget Application-Based Deployment for the actual execution of the deployment of the Cloud MTA Build tool will be used ([cloud mta build tool](#)). `cds add mta`

To understand the Multitarget Application Project, we will deep dive into the next subject: the MTA Development Project.

The Multitarget Application (MTA) Development Project

Recent trends and progress for programming languages, software design architectures such as micro-services, protocols like OData, and the diversity of multitiered and distributed deployment platforms have accelerated the trend towards applications constructed out of smaller, decoupled, and diverse modules.

Today, business applications are composed of multiple parts that are developed using different languages and technologies and deployed to various target runtime environments. This diversity introduces many life cycle challenges.

A Multitarget Application (MTA) is composed of multiple parts (modules), created with different technologies and deployed to different targets, but with a single, common life cycle.

Simply put, an MTA is logically a single application, consisting of multiple related and interdependent parts (herein called modules) that are developed using different technologies or programming paradigms and designed to run on different target runtime environments, with a single, consistent life cycle.

Development tools provided by SAP allow you to manage multiple applications (as "modules") in a unique development project, and deploy them via a unique archive file (MTA archive (MTAR)).

In SAP Business Application Studio, we have options to either create a Basic MTA project or an SAP Cloud Application Programming Model (CAP) project. Throughout this course, we developed the business application and the related artifacts using the project of type CAP having the following folder structure:

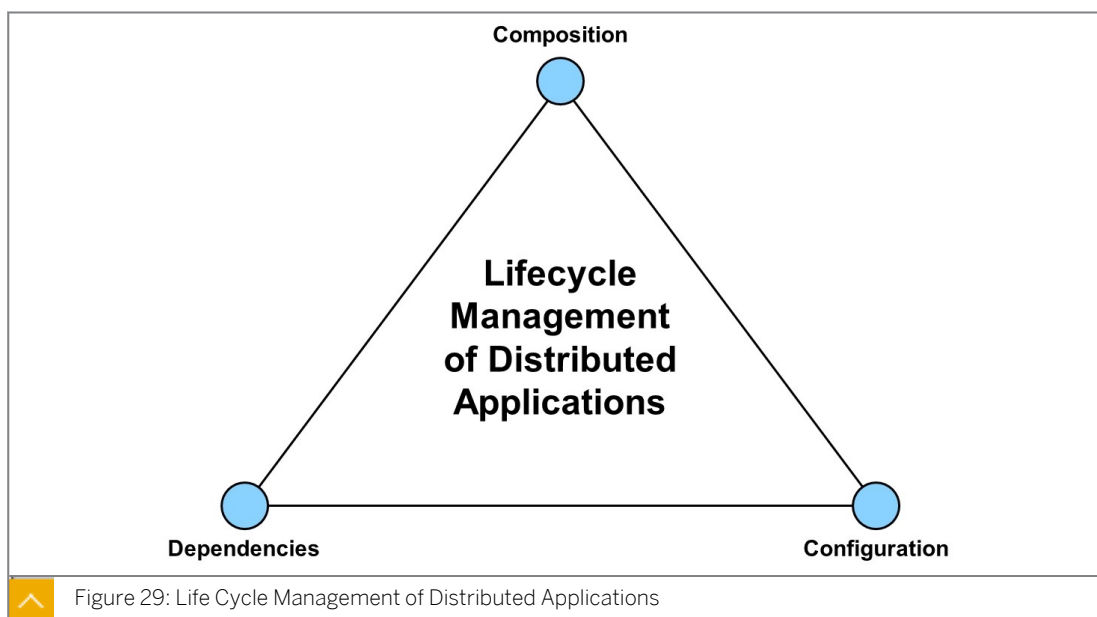
- db : For the database level schema model
- srv : For the service definition layer
- app : For UI artifacts

Describing the MTA Development Descriptor File

An MTA model is a platform-independent description of the different application modules, their interdependencies, and configuration data that they expose, and the resources they require to run. This model is specified, using [YAML](#), in descriptor files that accompany the development and deployment processes.

An MTA model serves the following purposes:

1. Define an application composed of multiple (heterogeneous) subcomponents (benefit: tools can establish a unique life cycle of these subcomponents).
2. Declare resources the application depends upon at runtime and/or deployment time (benefit: tools can allocate and bind such resources).
3. Define configuration variables (and their relation), whose values distinguish different deployments of the application (benefit: tools can bind subcomponents, can automate deployment based on default settings, or request missing mandatory values interactively).



As seen in the image, the application developer uses development tools to create the modules of an MTA and the corresponding MTA descriptor (`mta.yaml`).

The application can then be distributed in the form of an MTA archive including the MTA deployment descriptor (`mtad.yaml`). This specification describes an archive format as a convenient distribution file. As indicated by the direct arrow from development tools to the deployer.

MTA deployers may also accept the pure contents of this format, namely the directory structure of files with a deployment descriptor (`mtad.yaml`).

An administrator optionally augments the MTA model in the deployment descriptor with an extension descriptor (`mtaext.yaml`), and uses the MTA deployer to orchestrate the actual deployment.

- **Global elements:** these include the application identifier and version, a description (optional), copyrights, author, and so on.
- **Modules:** Modules created in the application, such as the SAP HANA database module and the Node.js module, including name, type, path, and requirements on other modules.
- **Resources:** A resource is something that is required by a module of the MTA at runtime, or at deployment time, but not provided inside the MTA. More precisely, an MTA descriptor declares a resource dependency, not the resource itself. Sometimes they are referred to as Backing Services, dependent services that are not provided by the application, such as XS Advanced User Account and Authentication (XSUAA), XS Advanced HANA Deployment Infrastructure (XSHDI) container, and XSJob-Scheduler.
- **Properties:** These can be specified when the value has to be determined during the deployment, for example, for generated URLs of other services or API keys.
- **Parameters:** Reserved variables of a module, which can be accessed by other modules, for example, user, app-name, default-host, or default-uri using the placeholder notation. Parameters can be read-only or read/write enabled.

4. Using Application Router

Its tasks are:

The App Router acts as a single point-of-entry gateway to route requests to. In particular, it ensures user login and authentication in combination with XSUAA. Its tasks are:

- The App Router dispatches requests to our backend microservices, thus acting as a reverse proxy. The backend microservices should not be directly accessible by the client.
- The App Router can serve static content such as web pages, SAPUI5, or another client-side code.
- The App Router manages the authentication flows for our entire application. For authentication (who the user is) and authorization (what the user is allowed to do), the App Router takes all incoming, unauthenticated requests and initiates an OAuth2 flow (authorization code grant) with the Extended Services for User Account and Authentication (XSUAA) service of the SAP BTP in the Cloud Foundry environment.

SAP offers two main options for using an approuter in the context of the Cloud Foundry environment: SAP Managed Approuter and Standalone Approuter.

SAP Managed Approuter for scenarios where a launchpad is the entry point to access your applications. The managed approuter functionality comes with the subscription to SAP Build Workzone. The applications deployed with the managed approuter is available in the HTML5 repository. Standalone App Router: for special scenarios where the approuter extensibility is mandatory or for building multitenant applications (SAAS) on SAP BTP. The application deployed with this custom approuter will be available in the CF space.

Use the following command to enhance the application configuration: `cds add approuter --for production`

Use the following command to enhance the application configuration: `cds add approuter --for production`.

5. Build the application

Run the following command to generate more deployment artifacts and prepare everything for production in a local `./gen` folder as a staging area. While `cds build` is included in the next step when running `mbt build`, you can also run it selectively as a test, and to inspect what is generated.

```
cds build --production
```

6. Deploy to Cloud Foundry

By running `mbt build`, an archive file (MTA archive (MTAR)) is created in the `gen` folder. To deploy the generated archive, you must run `cf deploy gen/mta.tar`. The url of the approuter application appears at the end in the log output.

Graphical MTA Editor

As an alternative to the code editor for the MTA file, the SAP Business Application Studio also provides the possibility to use the graphical MTA editor to make changes. You can find the form-based editor on the context menu of the `mta.yaml` file.

Summary

To deploy a CAP application to Cloud Foundry, there are specific steps that must be ensured: the BTP Cloud Foundry environment is properly setup, the CAP application is prepared, and the MTA descriptor file is configured by keeping in mind all the necessary bindings. With this, the application is successfully built and deployed.

Further Reading

Example:

- [Multitarget applications in CF environment](#)
- [CAP deployment](#)



LESSON SUMMARY

You should now be able to:

- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file

Preparing for Deployment in the SAP BTP, Cloud Foundry Runtime



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Prepare the application for deployment



LESSON SUMMARY

You should now be able to:

- Prepare the application for deployment

Deploying the Application



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Deploying the application



LESSON SUMMARY

You should now be able to:

- Deploying the application

Learning Assessment

1. Which statements are true about CAP's deployment options?

Choose the correct answers.

- ☐ A CAP applications can only be deployed on SAP BTP environments.
- ☐ B CAP has built-in support for multitenancy.
- ☐ C CAP can only be used for single tenant applications.
- ☐ D CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments.

2. What is a Multitarget application (MTA) comprised of?

Choose the correct answer.

- ☐ A Modules and resources
- ☐ B Modules and services
- ☐ C Modules and instances

3. *Choose the correct answer.*

4. After you run the command `cds add hana`, which file is updated with the required configuration?

Choose the correct answer.

- ☐ A package.js
- ☐ B package.cds
- ☐ C package.json
- ☐ D package.mta

5. What are the advantages of using an MTA file for deployment? Choose the two correct answers.

Choose the correct answers.

- ☐ A It supports red - green deployment.
- ☐ B It supports blue-green deployment.
- ☐ C It provides workflows.
- ☐ D It provides a build tool.

6. What are yaml files used for?

Choose the correct answer.

- ☐ A To describe documents
- ☐ B To describe data

7. Which statements about YAML files are correct? Choose the two correct answers.

Choose the correct answers.

- ☐ A YAML uses whitespace indentation for structuring purposes.
- ☐ B YAML uses tab indentation for structuring purposes.
- ☐ C YAML uses hyphens: - for comments.
- ☐ D YAML uses hashes: # for comments.

8. *Choose the correct answer.*

Learning Assessment - Answers

1. Which statements are true about CAP's deployment options?

Choose the correct answers.

- ☐ A CAP applications can only be deployed on SAP BTP environments.
- ☒ B CAP has built-in support for multitenancy.
- ☐ C CAP can only be used for single tenant applications.
- ☒ D CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments.

Correct. The statements: "CAP has built-in support for multitenancy" and "CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments" are correct.

2. What is a Multitarget application (MTA) comprised of?

Choose the correct answer.

- ☒ A Modules and resources
- ☐ B Modules and services
- ☐ C Modules and instances

Correct. The multitarget application (MTA) comprises modules and resources.

3. *Choose the correct answer.*

4. After you run the command `cds add hana`, which file is updated with the required configuration?

Choose the correct answer.

- ☐ A package.js
- ☐ B package.cds
- ☒ C package.json
- ☐ D package.mta

Correct. The package.json file is updated with the required configuration.

5. What are the advantages of using an MTA file for deployment? Choose the two correct answers.

Choose the correct answers.

- ☐ A It supports red - green deployment.
- ☒ B It supports blue-green deployment.
- ☐ C It provides workflows.
- ☒ D It provides a build tool.

Correct. The advantages of using an MTA file for deployment are: "it supports blue-green deployment", and "it provides a build tool".

6. What are yaml files used for?

Choose the correct answer.

- ☐ A To describe documents
- ☒ B To describe data

Correct. Yaml files are used for describing data.

7. Which statements about YAML files are correct? Choose the two correct answers.

Choose the correct answers.

- ☒ A YAML uses whitespace indentation for structuring purposes.
- ☐ B YAML uses tab indentation for structuring purposes.
- ☐ C YAML uses hyphens: - for comments.
- ☒ D YAML uses hashes: # for comments.

Correct. The following statements are correct: "YAML uses whitespace indentation for structuring purposes", and "YAML uses hashes: # for comments".

8. *Choose the correct answer.*

UNIT 8

Integrating into SAP Build Work Zone, Standard Edition

Lesson 1

Explaining SAP Build Work Zone, Standard Edition

129

Lesson 2

Integrating the App with SAP Build Work Zone

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UNIT OBJECTIVES

- Describe use case for SAP Build Work Zone, standard edition
- Integrate the application with SAP Build Work Zone, Standard Edition

Explaining SAP Build Work Zone, Standard Edition



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe use case for SAP Build Work Zone, standard edition

SAP Build Work Zone, Standard Edition

Introduction

In today's world, your company must develop cloud-based applications that will be used across various devices, such as mobile phones, tablets, and laptops. Ensuring a consistent look and feel across these devices is crucial to reduce the learning curve for your end users. Therefore, you need an efficient way to design the user interfaces of these applications, ensuring uniformity and ease-of-use regardless of the device.

SAP Build Work Zone, standard edition

In the modern business landscape, organizations often grapple with complex enterprise environments. These environments can be a mix of various content types, UI tools, IT systems, content repositories, applications, and channels. This complexity can pose a challenge for business users when they're trying to find the information they need.

Employees often express difficulty in switching between applications and feel overwhelmed when trying to locate the right application for a task. SAP Build Work Zone addresses these challenges by offering a centralized access point for business applications and information on any device.

What is the SAP Build Work Zone? SAP Build Work Zone is a tool that allows you to create business sites with ease. These sites provide centralized access to business applications and information. With prebuilt templates, user interface cards, and content widgets, you can visually build and manage sites without needing extensive technical knowledge.

Key Benefits

- **Centralized Access:** SAP Build Work Zone provides a single entry point for all your enterprise applications. This makes it easier to find and use the tools you need.
- **Improved Productivity:** It integrates with both cloud and on-premise systems. This streamlines your daily tasks and boosts efficiency.

Features of SAP Build Work Zone, standard edition

1. **Application Integration Central Access:** It offers a central point of access for apps and tasks. It integrates seamlessly with various UI technologies and third-party applications. Enhanced Productivity: It reduces the need to switch between different applications, saving time and effort.

2. **Intuitive and Engaging User Experience Personalized Sites:** It provides role-based, personalized business sites with flexible content structures and designs. User-Friendly Design: It ensures that users have an engaging and easy-to-navigate interface.
3. **Integrations SAP BTP Services:** It integrates with central SAP Business Technology Platform services, such as the inbox and cloud identity services. Third-Party Integrations: It allows integration with other third-party services, enhancing the functionality of your business sites.
4. **Extensibility Customization:** It offers an extensibility framework for customizations using shell plugins and custom branding. Flexible Design: It supports a variety of customizations, including translation, branding, shell plug-ins, and domain URLs.

Getting Started

To get started with SAP Build Work Zone, follow these steps:

1. **Create a Subscription:** Start by creating a subscription to SAP Build Work Zone, standard edition.
2. **Set Up Your Site:** Use prebuilt templates, user interface cards, and content widgets to set up your business site.
3. **Integrate Applications:** Integrate your enterprise applications and third-party services to provide a centralized access point.
4. **Customize Your Site:** Use the extensibility framework to customize your site. This includes branding and additional functionality.

Summary

SAP Build Work Zone, standard edition is a powerful tool that can help you simplify your enterprise landscape and improve productivity. By providing a centralized access point for all your applications and information, it makes it easier for business users to find what they need and complete their tasks efficiently.

Further Reading

- [What Is SAP Build Work Zone, standard edition?](#)
- [SAP Build Work Zone, standard edition](#)



LESSON SUMMARY

You should now be able to:

- Describe use case for SAP Build Work Zone, standard edition

Integrating the App with SAP Build Work Zone



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Integrate the application with SAP Build Work Zone, Standard Edition



LESSON SUMMARY

You should now be able to:

- Integrate the application with SAP Build Work Zone, Standard Edition

Learning Assessment

1. What is the primary benefit of using SAP Build Work Zone, standard edition?

Choose the correct answer.

- ☐ A It offers a complex and customizable coding environment.
- ☐ B It provides a centralized access point for business applications and information on any device.
- ☐ C It exclusively supports cloud-based solutions without on-premise integration.
- ☐ D It requires extensive technical knowledge to set up and manage.

2. Which feature of SAP Build Work Zone, standard edition allows for customizations using shell plugins and custom branding?

Choose the correct answer.

- ☐ A Application Integration
- ☐ B Extensibility
- ☐ C Intuitive and Engaging User Experience
- ☐ D Integrations

3. Which role collection is required in the Integrate with SAP Build Work Zone, Standard Edition?

Choose the correct answer.

- ☐ A SAP_ALL
- ☐ B Admin_All
- ☐ C Launchpad_Admin

Learning Assessment - Answers

1. What is the primary benefit of using SAP Build Work Zone, standard edition?

Choose the correct answer.

- ☐ A It offers a complex and customizable coding environment.
- ☒ B It provides a centralized access point for business applications and information on any device.
- ☐ C It exclusively supports cloud-based solutions without on-premise integration.
- ☐ D It requires extensive technical knowledge to set up and manage.

Correct. It provides a centralized access point for business applications and information on any device.

2. Which feature of SAP Build Work Zone, standard edition allows for customizations using shell plugins and custom branding?

Choose the correct answer.

- ☐ A Application Integration
- ☒ B Extensibility
- ☐ C Intuitive and Engaging User Experience
- ☐ D Integrations

Correct. Extensibility allows for customizations using shell plugins and custom branding.

3. Which role collection is required in the Integrate with SAP Build Work Zone, Standard Edition?

Choose the correct answer.

- ☐ A SAP_ALL
- ☐ B Admin_All
- ☒ C Launchpad_Admin

Correct. You need to assign your user to the Launchpad_Admin role collection.